

The Digital Gold Debate: Safe-Haven Properties and Dynamic Spillovers between Bitcoin, Gold, and the VN-Index

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Abstract

This study revisits the “Digital Gold” debate by examining whether Bitcoin can perform a safe-haven role similar to gold in the Vietnamese financial market. Using daily data from January 2015 to October 2024, the analysis focuses on Bitcoin, domestic gold prices, and the Vietnamese stock market represented by the VN-Index, which is treated as a proxy for the HOSE index. The empirical framework combines Johansen cointegration tests, a Vector Error Correction Model, Granger causality tests, generalized impulse response functions, forecast error variance decomposition, safe-haven stress regressions, EWMA dynamic correlations, and the Diebold–Yilmaz spillover index. The results show that Bitcoin, gold, and the VN-Index share a long-run equilibrium relationship. However, gold and Bitcoin play markedly different roles within this system. Gold exhibits a negative long-run association with Vietnamese equities and provides partial evidence of a weak safe-haven role, especially during moderate market stress. By contrast, Bitcoin shows a positive relationship with the VN-Index, has stronger short-run predictive power for equity movements, and tends to act as a net transmitter of shocks. The EWMA and stress-test results further indicate that Bitcoin moves more closely with the Vietnamese stock market during crisis periods rather than insulating investors from losses. Overall, gold remains the more reliable defensive asset for portfolio risk management.

1 Introduction

The rapid development of cryptocurrency markets has renewed the debate over whether Bitcoin can be regarded as “digital gold”. This debate is particularly relevant in emerging markets, where investors often face higher macroeconomic uncertainty, more volatile equity returns, and a narrower set of conventional hedging instruments. Gold has traditionally been viewed as a defensive asset, while Bitcoin has increasingly attracted attention as a decentralized and scarce digital asset. However, whether Bitcoin can provide the same safe-haven function as gold remains an empirical question, especially in country-specific markets such as Vietnam.

This study examines the safe-haven properties and dynamic spillovers among Bitcoin, gold, and the Vietnamese stock market. The Vietnamese equity market is represented by the VN-Index, which is the benchmark index associated with the Ho Chi Minh Stock Exchange. In this paper, the terms VN-Index and HOSE index are used interchangeably, as the VN-Index is commonly used to proxy the price movement of stocks listed on the Ho Chi Minh Stock Exchange. HOSE-related index documents also describe its index system as a market-capitalization-weighted set of indices based on companies listed on the Ho Chi Minh Stock Exchange (Ho Chi Minh Stock Exchange, 2025). Therefore, the VN-Index provides a suitable market-wide indicator for analysing the interaction between domestic equities and alternative assets.

Gold occupies a special position in both global finance and Vietnamese investment behaviour. In the international literature, gold is frequently analysed as a hedge or safe-haven asset. Baur and Lucey (2010) define a safe haven as an asset that is uncorrelated or negatively correlated with another asset during periods of market stress, and their results suggest that gold can act as a safe haven in extreme stock market conditions. Similarly, Baur and McDermott (2010) provide international evidence that gold may serve as both a hedge and a safe haven for several major equity markets, although this role can vary across countries and market conditions. In Vietnam, the gold market is also shaped by a distinctive regulatory environment. Decree

No. 24/2012/ND-CP placed gold trading activities under strong state management, including state control over gold-bar production and the import-export of gold materials for gold-bar production (Government of Vietnam, 2012). Although Decree No. 232/2025/ND-CP later amended this framework and removed several monopoly elements, gold trading remains subject to licensing, supervision, and macro-financial management by the authorities (Government of Vietnam, 2025a). This institutional background makes gold a particularly important asset to examine in the Vietnamese context.

Bitcoin, by contrast, represents a newer and more controversial asset class. Its limited supply, decentralized design, and independence from traditional monetary authorities have led some investors and researchers to compare it with gold (Henriques and Sadorsky, 2018). Nevertheless, Bitcoin differs substantially from gold in terms of volatility, market maturity, regulatory treatment, and investor behaviour. In Vietnam, cryptocurrencies were not banned outright, but for much of the sample period they were not clearly recognized as legal assets or lawful payment instruments. The Government News Portal noted in 2025 that Vietnam still lacked a clear definition of virtual currencies and digital assets, and that cryptocurrencies were not yet recognized as assets at that time (Viet Nam Government News, 2025). More recently, however, Vietnam has moved toward constructing a formal legal framework for digital and crypto assets. The Law on Digital Technology Industry, approved in 2025 and scheduled to take effect in 2026, introduces a legal framework for digital assets, including tokenized and virtual assets (National Assembly of Vietnam, 2025). In addition, Resolution No. 05/2025/NQ-CP establishes a pilot framework for the crypto-asset market in Vietnam (Government of Vietnam, 2025b). These developments make the question of Bitcoin’s financial role increasingly relevant for investors, regulators, and portfolio managers.

Against this background, this study contributes to the literature by comparing Bitcoin and gold as potential safe-haven assets for the Vietnamese stock market. Methodologically, the paper combines long-run and short-run time-series approaches. First, the Vector Error Correction Model is used to examine whether Bitcoin, gold, and the VN-Index share a long-run equilibrium relationship. Second, Granger causality tests, generalized impulse response functions, and forecast error variance decomposition are employed to study short-run dynamics. Third, safe-haven stress tests and EWMA rolling correlations are used to evaluate whether Bitcoin and gold behave defensively during periods of market stress. Finally, the Diebold–Yilmaz spillover framework is applied to identify the direction and intensity of risk transmission among the three markets (Diebold and Yilmaz, 2012a).

The empirical results suggest that the “digital gold” label should be treated with caution in the Vietnamese market. Gold appears to retain some defensive characteristics, especially during moderate equity-market stress, and its relationship with the VN-Index is broadly consistent with its traditional role as a diversifier and weak safe haven. Bitcoin, however, shows a closer association with risk-asset behaviour. It tends to move more closely with the VN-Index in stressed periods and appears to transmit shocks to the Vietnamese equity market rather than absorb them. These findings imply that Bitcoin may not yet provide the same portfolio protection as gold for Vietnamese equity investors. Therefore, the study offers relevant implications for portfolio allocation, risk management, and the ongoing policy discussion on crypto-asset regulation in Vietnam.

2 Literature Review

The question of whether Bitcoin can be regarded as “digital gold” has attracted growing attention in financial economics, especially after the rapid expansion of cryptocurrency markets and the increasing demand for alternative assets during periods of financial uncertainty. Gold has traditionally been considered a store of value, an inflation hedge, and a safe-haven asset. Bitcoin, by contrast, is often compared with gold because of its limited supply, decentralized design, and potential role as an alternative investment asset. However, the empirical literature has not reached a consensus on whether Bitcoin possesses the same hedging and safe-haven characteristics as gold.

A first strand of literature directly examines the “digital gold” hypothesis by comparing Bitcoin and gold in terms of volatility, dynamic correlation, and portfolio performance. Klein et al. (2018) argue that Bitcoin is not the new gold, showing that Bitcoin and gold exhibit substantially different risk-return characteristics. Their findings suggest that Bitcoin is much more volatile than gold and does not provide stable safe-haven properties during periods of financial distress. Similarly, Choi and Shin (2022) distinguish between Bitcoin’s

role as an inflation hedge and its role as a safe haven. Their results indicate that although Bitcoin may respond to inflation-related shocks, this does not necessarily imply that it can protect investors during stock market turmoil. This distinction is important because an asset may hedge against inflation without serving as a safe haven in crisis periods. From a portfolio perspective, Henriques and Sadorsky (2018) investigate whether Bitcoin can replace gold in an investment portfolio using DCC, ADCC, and GO-GARCH models. They show that Bitcoin may improve certain portfolio outcomes, but portfolio performance alone does not prove that Bitcoin has the same safe-haven function as gold.

A second strand of literature focuses on the conceptual distinction among hedge, diversifier, and safe-haven assets. Following the framework of Baur and Lucey (2010), a hedge is defined as an asset that is uncorrelated or negatively correlated with another asset on average, while a diversifier is positively but imperfectly correlated with another asset. A safe haven is more restrictive, referring to an asset that is uncorrelated or negatively correlated with another asset during periods of market stress or financial turmoil. Kumar and Padakandla (2022) apply this framework to compare the safe-haven properties of gold and Bitcoin. Their findings suggest that gold provides more consistent safe-haven benefits, whereas the evidence for Bitcoin is mixed and depends strongly on market conditions. This conceptual distinction is particularly relevant to the present study because the objective is not simply to examine whether Bitcoin or gold generates positive returns, but whether these assets behave defensively when the Vietnamese stock market experiences sharp declines.

A third strand of literature examines dynamic relationships and spillover effects among Bitcoin, gold, commodities, and equity markets. These studies emphasize that cross-asset relationships are time-varying and may change substantially across normal and crisis periods. Choi and Shin (2022) employ a VAR framework to examine the responses of Bitcoin and gold to shocks related to inflation, uncertainty, and financial market conditions. This approach provides a useful methodological foundation for analysing shock transmission through Granger causality, impulse response functions, and forecast error variance decomposition. Extending this logic, Wen et al. (2022) use a TVP-VAR model to compare the safe-haven ability of gold and Bitcoin during the COVID-19 pandemic. Their results show that gold can serve as a short-term safe haven for stock markets, while Bitcoin does not display the same defensive capacity. Moreover, the safe-haven role of gold becomes stronger when the pandemic shock intensifies.

In a similar direction, Wang et al. (2022) investigate the safe-haven ability of Bitcoin, gold, and commodities using a TVP-VAR spillover index. Their findings reveal that spillover effects are highly time-varying and tend to increase sharply after extreme events. They also show that commodities and gold generally perform better than Bitcoin in terms of safe-haven ability. This evidence is especially relevant for studies that incorporate dynamic spillover analysis, rolling-window estimation, or time-varying connectedness measures. Bouri et al. (2018) further show that spillovers between Bitcoin and other assets can differ between bull and bear market regimes, implying that nonlinearities, asymmetries, and structural breaks should be considered when evaluating Bitcoin’s hedging and safe-haven roles. Overall, this literature supports the use of VAR/VECM-based models, rolling VAR, DCC-GARCH, and stress-period analysis to capture changing relationships among Bitcoin, gold, and stock markets.

A fourth strand of literature focuses on Asian, ASEAN, and emerging markets. This is particularly important because empirical results from developed markets such as the United States or G7 economies may not be directly applicable to Vietnam. Emerging markets differ in terms of investor structure, market liquidity, exchange-rate exposure, regulatory environment, and financial market development. Sinlapates et al. (2023) provide important regional evidence by examining the hedge and safe-haven roles of Bitcoin and gold for ASEAN+6 equity markets using multivariate GARCH models. Their findings suggest that Bitcoin and gold do not consistently act as safe havens for most ASEAN+6 stock markets, and in some cases may even increase downside risk. This result challenges the simple assumption that gold and Bitcoin always protect investors during periods of market stress. Wang et al. (2019) provide further evidence from China by applying a VAR-GARCH-BEKK framework to examine mean and volatility spillovers between Bitcoin and major Chinese financial assets, including stocks, gold, foreign exchange, monetary assets, and bonds. Their study indicates that Bitcoin’s hedging and safe-haven properties are market-specific and depend on the structure of the domestic financial system.

Despite the growing literature on Bitcoin, gold, and financial market spillovers, there remains a clear research gap in the Vietnamese context. Existing studies mainly focus on global markets, G7 economies, China, or broader ASEAN+6 samples. Vietnam-specific evidence remains limited, particularly when using

the VN-Index, which is treated in this study as equivalent to the HOSE index, together with domestic gold prices and Bitcoin converted into Vietnamese dong. This distinction is important because the Vietnamese gold market is influenced by domestic regulation and may not move identically with international gold prices. At the same time, Bitcoin has not yet been formally recognized as a conventional financial asset in Vietnam, which may affect its investment role and its interaction with traditional assets.

Therefore, this study contributes to the literature by examining whether Bitcoin can perform a role similar to gold in the Vietnamese financial market. Specifically, it investigates the dynamic relationship among Bitcoin, gold, and the HOSE index through both baseline econometric models and extended time-varying approaches. The baseline analysis includes unit root tests, Johansen cointegration tests, VAR/VECM estimation, Granger causality tests, impulse response functions, and forecast error variance decomposition. The extended analysis considers dynamic correlations, rolling-window relationships, and stress-period behavior to evaluate whether Bitcoin and gold act as hedge, diversifier, or safe-haven assets during Vietnamese stock market downturns. By focusing on Bitcoin/VND, domestic gold prices, and the HOSE index, this study provides localized evidence to the broader digital gold debate and offers new insights into the role of alternative assets in an emerging market setting.

3 Methodology

3.1 Data and Preliminary Processing

Table 1: Descriptive statistics of price variables

(a) Daily data						(b) Weekly data					
	VN_Index	Gold_Price	BTC_VND	BTC_USD	USD_VND		VN_Index	Gold_Price	BTC_VND	BTC_USD	USD_VND
count	2246.00	2246.00	2246.00	2246.00	2246.00	count	509.00	509.00	509.00	509.00	509.00
mean	969.62	49699960.00	439858000.00	18562.39	23122.89	mean	975.38	50071180.00	451295300.00	19025.31	23136.40
std	259.58	15616480.00	484310200.00	20046.70	819.62	std	260.39	15858580.00	491484500.00	20304.23	830.99
min	521.88	32450000.00	3903992.00	183.00	21115.00	min	522.24	32450000.00	4405810.00	206.70	21232.00
25%	742.12	36420000.00	54183490.00	2394.65	22700.00	25%	749.72	36420000.00	56292510.00	2479.30	22700.00
50%	976.06	41795000.00	217229100.00	9343.45	23108.00	50%	982.90	42050000.00	219441900.00	9426.20	23124.00
75%	1174.24	66300000.00	696876200.00	29638.75	23374.50	75%	1179.25	66500000.00	717005100.00	30343.00	23393.00
max	1528.57	90000000.00	1838643000.00	72974.00	25469.00	max	1528.57	89000000.00	1776456000.00	70688.00	25465.00

The dataset used in this study covers the period from January 1, 2015 to October 31, 2024, including VN-Index, domestic gold prices, BTC/USD prices, and the USD/VND exchange rate. The data were collected from publicly available financial data sources, namely vn.investing.com and giavang.org. To ensure consistency across assets, the observations were aligned according to the trading days of the Vietnamese stock market. Since Bitcoin is not officially recognized as a tradable electronic asset in Vietnam, BTC/USD prices were converted into BTC/VND using the corresponding USD/VND exchange rate. Accordingly, both gold and Bitcoin prices are expressed in Vietnamese dong, while the VN-Index is measured in index points.

3.1.1 Return Computation

Let P_t denote the closing price of asset i at time t . All price series are transformed into continuously compounded (log) returns:

$$r_t = \ln P_t - \ln P_{t-1} = \Delta \ln P_t, \quad (1)$$

where $\ln P_t$ is the natural logarithm of the price level. Log returns are preferred over simple returns because they are time-additive, approximately normally distributed for small changes, and facilitate stationarity in empirical models (see, e.g., Campbell et al., 1997).

3.1.2 Descriptive Statistics

For each return series we report the sample mean (μ), standard deviation (σ), skewness ($\mathcal{S}$), excess kurtosis ($\mathcal{K}$), and the Jarque–Bera (JB) normality statistic (Jarque and Bera, 1987):

$$\text{JB} = \frac{T}{6} \left[\mathcal{S}^2 + \frac{(\mathcal{K} - 3)^2}{4} \right] \sim \chi^2(2) \quad \text{under } H_0: \text{normality}, \quad (2)$$

where T is the number of observations. We also compute pairwise Pearson correlations among all return series as a first-pass assessment of co-movement.

3.1.3 Unit Root Tests

Prior to cointegration analysis, we test whether each log-price series $\ln P_t$ contains a unit root (i.e. is integrated of order one, $I(1)$) using three complementary procedures to ensure robustness across different assumptions about the data-generating process.

Augmented Dickey–Fuller (ADF) test. Dickey and Fuller (1979) propose the regression

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{j=1}^p \delta_j \Delta y_{t-j} + \varepsilon_t, \quad (3)$$

where $y_t = \ln P_t$, α is a constant, βt is an optional time trend, and p lagged differences are included to whiten the residuals $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$. The null hypothesis is $H_0: \gamma = 0$ (unit root); the optimal lag length p is selected by the Akaike Information Criterion (AIC).

Phillips–Perron (PP) test. Phillips and Perron (1988) adopt the same null hypothesis but use a non-parametric correction for serial correlation and heteroskedasticity in ε_t , making the test robust to a wider class of error processes without requiring explicit lag augmentation.

KPSS test. Kwiatkowski et al. (1992) reverse the null, testing H_0 : stationarity against H_1 : unit root via the statistic

$$\text{KPSS} = \frac{1}{T^2} \frac{\sum_{t=1}^T S_t^2}{\hat{\sigma}_\ell^2}, \quad (4)$$

where $S_t = \sum_{s=1}^t \hat{e}_s$ is the partial-sum process of OLS residuals from a regression of y_t on a constant (and trend), and $\hat{\sigma}_\ell^2$ is a Newey–West long-run variance estimator with bandwidth ℓ . Combining ADF/PP (null: unit root) with KPSS (null: stationarity) provides a cross-validation strategy: a series is classified as $I(1)$ only when ADF and PP fail to reject their null *and* KPSS rejects its null at the same significance level.

A series is confirmed as $I(1)$ if the unit-root null cannot be rejected in *levels* but is rejected in *first differences*. Only if all series share the same order of integration does the analysis proceed to the Johansen cointegration test (Section 3.2.1).

3.2 Cointegration and VECM Estimation

3.2.1 Johansen Cointegration Test

Given that there exist series confirmed to be integrated of order one $I(1)$ in Section 3.1.3, we apply the Johansen (1988) maximum likelihood framework to test for the number of cointegrating vectors r among n variables. In this study, the vector of endogenous variables consists of three log-price series:

$$\mathbf{y}_t = [\ln P_t^{\text{HOSE}} \quad \ln P_t^{\text{Gold}} \quad \ln P_t^{\text{BTC}}]^\top, \quad (5)$$

where $n = 3$. Consider a p -th order vector autoregression (VAR) in levels:

$$\mathbf{y}_t = \boldsymbol{\mu} + \sum_{i=1}^p \mathbf{A}_i \mathbf{y}_{t-i} + \boldsymbol{\varepsilon}_t, \quad (6)$$

where $\mathbf{y}_t$ is an $(n \times 1)$ vector of log-price series, $\boldsymbol{\mu}$ is a constant vector, $\mathbf{A}_i$ are $(n \times n)$ coefficient matrices, and $\boldsymbol{\varepsilon}_t$ is a vector of innovations.

Following Johansen (1991), the VAR in (6) can be re-parameterised into the following vector error-correction form:

$$\Delta \mathbf{y}_t = \boldsymbol{\alpha} \left(\boldsymbol{\beta}^\top \mathbf{y}_{t-1} + c \right) + \sum_{i=1}^{k_\Delta} \boldsymbol{\Gamma}_i \Delta \mathbf{y}_{t-i} + \boldsymbol{\varepsilon}_t, \quad (7)$$

where $k_\Delta = p-1$ denotes the number of lagged differences in the VECM representation. The term c represents a restricted constant in the cointegrating relation, which is consistent with the deterministic specification used in the empirical implementation.

The long-run impact matrix is given by:

$$\boldsymbol{\Pi} = \boldsymbol{\alpha} \boldsymbol{\beta}', \quad (8)$$

where $\boldsymbol{\Pi}$ is of rank r , $\boldsymbol{\beta}$ contains the r cointegrating vectors, and $\boldsymbol{\alpha}$ contains the adjustment coefficients. If $r = 0$, no cointegrating relationship exists among the variables. If $0 < r < n$, the variables share one or more long-run equilibrium relationships and a VECM specification is appropriate.

The number of lagged differences, k_Δ , is selected prior to the cointegration test using the Hannan–Quinn information criterion (HQIC) with a maximum lag length of 10. This corresponds to selecting `k_ar_diff` in the VECM framework. The deterministic component is specified as `deterministic = "ci"`, implying that the constant is restricted to the cointegrating equation rather than entering the short-run dynamics unrestrictedly.

The cointegration rank r is then determined using the Johansen trace statistic at the 5% significance level. The trace statistic tests

$$H_0: \text{rank}(\boldsymbol{\Pi}) \leq r$$

against

$$H_1: \text{rank}(\boldsymbol{\Pi}) > r.$$

It is computed as

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_i), \quad (9)$$

where $\hat{\lambda}_i$ are the ordered eigenvalues obtained from the Johansen maximum likelihood procedure after accounting for the lagged first differences. Critical values follow the non-standard asymptotic distributions tabulated by Osterwald-Lenum (1992).

3.2.2 VECM Estimation

If the Johansen test confirms $r \geq 1$ cointegrating relationships, we estimate the full VECM of order $p-1$:

$$\Delta \mathbf{y}_t = \boldsymbol{\mu} + \underbrace{\boldsymbol{\alpha} \boldsymbol{\beta}' \mathbf{y}_{t-1}}_{\text{ECT}} + \sum_{i=1}^{p-1} \boldsymbol{\Gamma}_i \Delta \mathbf{y}_{t-i} + \boldsymbol{\varepsilon}_t, \quad (10)$$

where the error-correction term (ECT), $\boldsymbol{\beta}' \mathbf{y}_{t-1}$, measures deviations from the long-run equilibrium, and $\boldsymbol{\alpha}$ governs the speed at which each variable adjusts back to equilibrium. A statistically significant and negative α_i confirms that variable i is endogenous and error-correcting.

3.2.3 VECM Diagnostics

Model adequacy is assessed through four diagnostic procedures applied to the VECM residuals $\hat{\boldsymbol{\varepsilon}}_t$:

1. **Stability**—All roots of the companion matrix must lie strictly inside the unit circle; r roots are exactly unity by construction (Lütkepohl, 2005).
2. **Serial correlation**—The multivariate Breusch–Godfrey LM test (Godfrey, 1978) evaluates H_0 : no autocorrelation up to lag h .

3. **Conditional heteroskedasticity**—The ARCH-LM test (Engle, 1982) is applied equation-by-equation to detect remaining volatility clustering in $\hat{\varepsilon}_{i,t}$.
4. **Normality**—The multivariate Jarque–Bera test (Jarque and Bera, 1987) assesses whether the residuals are jointly normally distributed.

Rejection of normality is common with financial return data and does not invalidate the VECM estimates, as the Johansen procedure retains consistency under non-normality (Gonzalo, 1994). Rejection of the LM or ARCH-LM tests, however, may indicate misspecification and warrants lag re-selection.

3.3 Long-run and Short-run Analysis

3.3.1 Long-run Analysis

The long-run equilibrium relationship is characterised by the normalised cointegrating vector β . Following standard practice, we normalise β on the HOSE index so that the cointegrating equation takes the form:

$$\ln P_t^{\text{HOSE}} = \beta_0 + \beta_1 \ln P_t^{\text{BTC}} + \beta_2 \ln P_t^{\text{Gold}} + u_t, \quad u_t \sim I(0), \quad (11)$$

where $u_t = \beta' \mathbf{y}_t$ is the stationary equilibrium error, whose lagged value u_{t-1} serves as the error correction term (ECT) in the short-run dynamic model. Statistical significance of β_1 and β_2 is assessed using standard asymptotic t -tests on the normalised coefficients. The adjustment vector α is reported alongside β to characterise the speed at which each asset absorbs disequilibria.

3.3.2 Short-run Dynamics

Short-run causal relationships are examined through three tools estimated within the VECM framework.

Block Granger causality. We test whether lags of $\Delta \mathbf{y}_{j,t}$ jointly improve the forecast of $\Delta \mathbf{y}_{i,t}$ via the restriction $H_0: \Gamma_{i \leftarrow j} = \mathbf{0}$, tested with an LR χ^2 statistic (Granger, 1969). This is applied both with and without the ECT to distinguish *short-run* from *long-run* causality.

Generalised Impulse Response Functions (GIRF).

Unlike orthogonalised IRFs, GIRFs (Pesaran and Shin, 1998) do not depend on the ordering of variables in the system:

$$\text{GIRF}(h, \delta_j, \Omega_{t-1}) = \frac{1}{\sqrt{\sigma_{jj}}} \mathbf{A}_h \Sigma \mathbf{e}_j, \quad (12)$$

where $\mathbf{A}_h$ is the h -step-ahead moving-average coefficient matrix, Σ is the residual covariance matrix, $\mathbf{e}_j$ is a selection vector for variable j , and σ_{jj} is the j -th diagonal element of Σ . GIRFs trace the dynamic response of each asset to a one-standard-deviation shock in another asset over a 20-trading-day horizon.

Confidence bands for the GIRFs are constructed using a residual-based bootstrap (Kilian, 1998; Benkwitz et al., 2001; Lütkepohl, 2005). At each of $B = 1,000$ bootstrap replications, we resample with replacement from the centred VECM residuals,

$$\{\hat{\varepsilon}_t - \bar{\varepsilon}\}_{t=1}^T.$$

A pseudo-sample of the same length T is then generated recursively using the estimated VECM coefficient matrices. The model is re-estimated for each pseudo-sample, and the GIRF is recomputed at each horizon h .¹

Generalised Forecast Error Variance Decomposition (GFEVD). The GFEVD (Pesaran and Shin, 1998) quantifies the proportion of the h -step forecast error variance of variable i attributable to shocks in variable j :

$$\theta_{ij}^g(h) = \frac{\sigma_{jj}^{-1} \sum_{l=0}^{h-1} (\mathbf{e}_i' \mathbf{A}_l \Sigma \mathbf{e}_j)^2}{\sum_{l=0}^{h-1} \mathbf{e}_i' \mathbf{A}_l \Sigma \mathbf{A}_l' \mathbf{e}_i}. \quad (13)$$

Because $\sum_j \theta_{ij}^g(h) \neq 1$ in general, each entry is normalised as $\tilde{\theta}_{ij}^g(h) = \theta_{ij}^g(h) / \sum_j \theta_{ij}^g(h)$.

¹The residual-based bootstrap is preferred over the asymptotic delta method because daily financial returns exhibit non-normality and conditional heteroskedasticity (Section 4.2.4), which can distort asymptotic standard errors for nonlinear functions of the VECM parameters (Benkwitz et al., 2001).

3.4 Safe-haven Stress Test

To assess whether Bitcoin (BTC) and gold serve as safe-haven assets for the Vietnamese stock market, this study builds upon the regression-based framework proposed by Baur and McDermott (2010). An asset is regarded as a safe haven if it is uncorrelated or negatively correlated with the stock market specifically during periods of extreme market stress.

Although the original framework is estimated using raw asset returns, applying such a specification directly to the present dataset may be econometrically inappropriate. As indicated by the Johansen cointegration test in the previous step, there exists a long-run equilibrium relationship among the VN-Index, gold, and Bitcoin (Johansen, 1988, 1991). Estimating an independent OLS regression on raw returns would therefore ignore the error-correction term (ECT), which may lead to omitted-variable bias. Moreover, the theoretical notion of a safe haven is concerned with an asset's ability to hedge against unexpected market panics rather than predictable long-run price dynamics.

To address this issue and isolate market innovations, the traditional safe-haven regression is modified by using the structural shocks extracted from the VECM residuals in 3.2.2. By filtering out short-run inertia and long-run adjustment dynamics, these residuals are intended to capture the unanticipated, panic-driven component of asset movements that is central to the safe-haven property. The following OLS regression is estimated:

$$\epsilon_{i,t} = \alpha + \beta_0 \epsilon_{\text{HOSE},t} + \beta_1 \epsilon_{\text{HOSE},t} D_{10\%,t} + \beta_2 \epsilon_{\text{HOSE},t} D_{5\%,t} + \beta_3 \epsilon_{\text{HOSE},t} D_{1\%,t} + \mu_{i,t}, \quad (14)$$

where $\epsilon_{i,t}$ denotes the VECM-extracted residual shock of asset i , with $i \in \{\text{BTC}, \text{Gold}\}$, and $\epsilon_{\text{HOSE},t}$ denotes the corresponding residual shock of the VN-Index at time t . The dummy variables $D_{q\%,t}$ represent increasingly severe stock-market downturns.

To capture the incremental effect of extreme market stress, these dummy variables are constructed in a cumulative, or nested, manner:

$$D_{10\%,t} = \begin{cases} 1, & \text{if } \epsilon_{\text{HOSE},t} \text{ lies below the 10th percentile of its empirical distribution,} \\ 0, & \text{otherwise,} \end{cases} \quad (15)$$

$$D_{5\%,t} = \begin{cases} 1, & \text{if } \epsilon_{\text{HOSE},t} \text{ lies below the 5th percentile of its empirical distribution,} \\ 0, & \text{otherwise,} \end{cases} \quad (16)$$

$$D_{1\%,t} = \begin{cases} 1, & \text{if } \epsilon_{\text{HOSE},t} \text{ lies below the 1st percentile of its empirical distribution,} \\ 0, & \text{otherwise.} \end{cases} \quad (17)$$

Under this cumulative specification, observations falling below the 5th percentile also belong to the 10% stress regime, while observations falling below the 1st percentile simultaneously belong to the 10%, 5%, and 1% stress regimes.

Accordingly, the total co-movement between asset i and the Vietnamese stock market at each stress level is obtained by summing the baseline coefficient and the relevant interaction coefficients. Specifically, the total effects are defined as follows:

$$\theta_{10\%} = \beta_0 + \beta_1, \quad (18)$$

$$\theta_{5\%} = \beta_0 + \beta_1 + \beta_2, \quad (19)$$

$$\theta_{1\%} = \beta_0 + \beta_1 + \beta_2 + \beta_3. \quad (20)$$

The safe-haven property is then evaluated based on these total effects. A weak safe haven is identified when the total co-movement is statistically indistinguishable from zero during market stress, whereas a strong safe haven is identified when the total co-movement is significantly negative.

Statistical inference is conducted using heteroskedasticity and autocorrelation consistent (HAC) standard errors, as proposed by Newey and West (1987). This approach is appropriate given the presence of volatility clustering and ARCH effects detected in the VECM diagnostics, as it provides more reliable inference in the presence of heteroskedasticity and serial correlation.

3.5 Dynamic Correlation Analysis via EWMA

While the safe-haven stress test in Section 1.4 provides a static assessment of co-movement during extreme market conditions, a fixed-window regression cannot capture how the relationship between assets evolves over time. To address this limitation, we complement the static framework with a dynamic correlation analysis using the **Exponentially Weighted Moving Average (EWMA)** approach, commonly known as the RiskMetrics methodology (J.P. Morgan and Reuters, 1996).

3.5.1 Construction of EWMA Dynamic Correlations

Rather than computing simple rolling correlations on raw returns—which conflates long-run equilibrium dynamics with short-run noise and suffers from the "ghosting effect" of equally weighted windows—we extract the VECM residuals $\epsilon_{i,t}$ from the estimated model. Using VECM residuals ensures that the measured co-movement is purged of the common stochastic trend and short-run autoregressive dynamics, thereby isolating idiosyncratic shock transmission between assets.

The EWMA method assigns exponentially declining weights to historical observations, meaning recent shocks have a greater impact on current volatility and correlation than older shocks. The conditional variance ($h_{ii,t}$) and covariance ($h_{ij,t}$) between asset i and j at time t are updated iteratively:

$$h_{ii,t} = (1 - \lambda)\epsilon_{i,t-1}^2 + \lambda h_{ii,t-1} \quad (21)$$

$$h_{ij,t} = (1 - \lambda)\epsilon_{i,t-1}\epsilon_{j,t-1} + \lambda h_{ij,t-1} \quad (22)$$

where λ is the decay factor. Following the standard RiskMetrics specification for daily financial data, we set $\lambda = 0.94$.

The dynamic conditional correlation $\rho_{ij,t}^{EWMA}$ is then computed as:

$$\rho_{ij,t}^{EWMA} = \frac{h_{ij,t}}{\sqrt{h_{ii,t}h_{jj,t}}} \quad (23)$$

This approach robustly handles the volatility clustering (ARCH effects) detected in our VECM diagnostics and provides a more responsive and smoother correlation series compared to standard rolling windows, avoiding convergence issues typical of DCC-GARCH models.

3.5.2 Safe-haven Identification via Dynamic Correlations

We focus on two dynamic correlation pairs: $\rho_{\text{BTC-HOSE},t}^{EWMA}$ and $\rho_{\text{Gold-HOSE},t}^{EWMA}$. An asset i is classified as a *time-varying safe haven* for the Vietnamese stock market at time t under a specific stress level if:

$$\rho_{i\text{-HOSE},t}^{EWMA} \leq 0 \quad \text{and} \quad \epsilon_{\text{HOSE},t} < Q_q(\epsilon_{\text{HOSE}}) \quad (24)$$

where Q_q denotes the q -th percentile of the unconditional residual distribution, with $q \in \{10\%, 5\%, 1\%\}$ capturing increasingly severe market downturns. This multi-threshold specification ensures consistency with the static stress levels defined in Section 3.4.

This condition operationalizes the Baur and McDermott (2010) definition in a dynamic setting: an asset serves as a safe haven only when its dynamic correlation with the distressed market turns non-positive during specific stress episodes, rather than on average across all market conditions.

3.6 Extension Analysis

To examine whether the long-run and short-run dynamics identified in Sections 3.2–3.3 are stable across different market regimes, we extend the baseline VECM framework along two dimensions.

3.6.1 Rolling VECM: Time-varying Cointegration and Causality

A single full-sample VECM implicitly assumes parameter stability over the entire estimation period. This assumption is potentially violated by the structural break induced by the COVID-19 pandemic, which caused synchronised global asset price dislocations (Akhtaruzzaman et al., 2021). To allow for time-varying parameters, we estimate the VECM recursively over a rolling window of W observations, re-estimating the full system including the Johansen rank test, cointegrating vectors β , adjustment coefficients α , and short-run matrices Γ_i at each step (Hansen, 1992).

From each rolling window estimate, we extract:

1. **Time-varying ECT coefficients $\hat{\alpha}_{i,t}$:** tracks how the speed of adjustment of each asset to the long-run equilibrium changes over time. A shift from significant to insignificant $\hat{\alpha}$ suggests a temporary breakdown in the cointegrating relationship.
2. **Time-varying Block Granger statistics:** The p -value of the block Granger χ^2 test (Section 3.3.2) is tracked at each rolling window to detect periods when the short-run causal link between BTC (or gold) and HOSE strengthens or dissipates.

Results are contrasted across three sub-periods: *Pre-COVID* (before 31 January 2020), *During-COVID* (1 February 2020 – 31 December 2021), and *Post-COVID* (from 1 January 2022 onwards), following the periodisation of Yarovaya et al. (2022). Sub-period breakdowns are reported in Appendix a4 alongside the rolling VECM figures in the main text.

3.6.2 Diebold–Yilmaz Spillover Index

To quantify the directional transmission of volatility shocks across assets, we implement the Diebold and Yilmaz (2012b) spillover framework, which decomposes the total forecast error variance of the system into *own*- and *cross-variable* contributions using the GFEVD defined in equation (13).

Total spillover index. The aggregate connectedness of the system at horizon H is:

$$S^g(H) = \frac{\sum_{i \neq j} \tilde{\theta}_{ij}^g(H)}{\sum_{i,j} \tilde{\theta}_{ij}^g(H)} \times 100, \quad (25)$$

where $\tilde{\theta}_{ij}^g(H)$ is the normalised GFEVD entry from equation (13).

Directional spillovers. Spillovers *transmitted from* variable j to all others:

$$S_{\bullet \leftarrow j}^g(H) = \frac{\sum_{i \neq j} \tilde{\theta}_{ij}^g(H)}{\sum_{i,j} \tilde{\theta}_{ij}^g(H)} \times 100. \quad (26)$$

Spillovers *received by* variable i from all others:

$$S_{i \leftarrow \bullet}^g(H) = \frac{\sum_{j \neq i} \tilde{\theta}_{ij}^g(H)}{\sum_{i,j} \tilde{\theta}_{ij}^g(H)} \times 100. \quad (27)$$

Net spillover and transmitter/receiver classification. The net spillover of variable j is:

$$S_j^{g,\text{net}}(H) = S_{\bullet \leftarrow j}^g(H) - S_{j \leftarrow \bullet}^g(H). \quad (28)$$

A variable is classified as a *net transmitter* of shocks if $S_j^{g,\text{net}}(H) > 0$, and a *net receiver* if $S_j^{g,\text{net}}(H) < 0$ (Diebold and Yilmaz, 2014). The forecast horizon is set to $H = 10$ trading days, following the convention in the connectedness literature (Diebold and Yilmaz, 2012b, 2014). Furthermore, an extended horizon of $H = 20$ is also specified to evaluate whether the spillover effects persist over a longer period.

4 Empirical Result

4.1 Preliminary Results

Figure 1 illustrates the time-series movements of the asset prices and index values over the sample period. The figure clearly captures several major episodes of market turbulence, particularly the sharp fluctuations during the COVID-19 period and the cryptocurrency market crash in 2022. On the other hand, the descriptive statistics reported in Table 2 provide strong evidence for the principle of “*high risk, high return*”. Among the three assets, Bitcoin exhibits the highest average return, but it is also associated with the greatest level of risk. This finding weakens the argument that Bitcoin can serve as a stable *safe-haven* asset in the same manner as gold.

In addition, the negative skewness values indicate that the return distributions of these assets tend to generate frequent small gains but are occasionally exposed to large negative shocks, reflecting substantial downside risk. This pattern is most pronounced for the VN-Index, which records the highest degree of negative skewness. The high kurtosis values further suggest that extreme events occur much more frequently than would be predicted under a normal distribution.

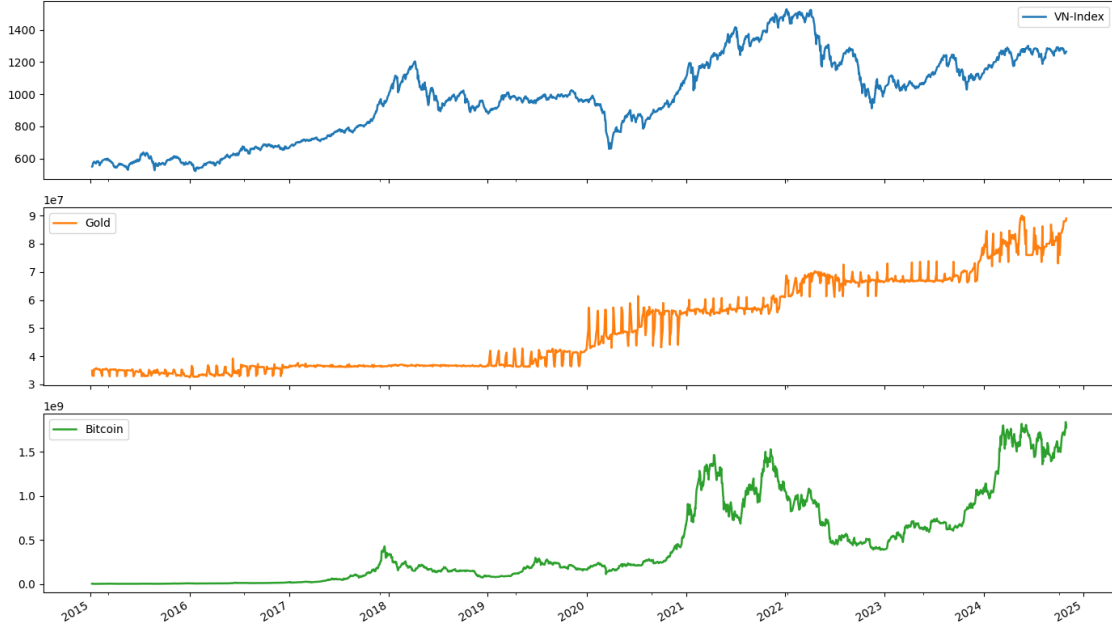


Figure 1: Price series chart of VN-Index, Gold, and Bitcoin

In particular, gold exhibits a kurtosis value above 20, implying that gold returns are usually concentrated around small fluctuations, but may react sharply in the tails when major macroeconomic shocks occur. Moreover, the Jarque–Bera test results show p-values close to zero, leading to a strong rejection of the null hypothesis of normality at the 1% significance level. Therefore, the return series of the VN-Index, gold, and Bitcoin do not follow a normal distribution. Finally, the correlation matrix reported in Table 3 shows that the correlations among the variables are relatively weak, suggesting limited linear co-movement between the Vietnamese stock market, gold, and Bitcoin during the sample period.

Table 2: Descriptive statistics and Jarque-Bera test of return series

	Mean	SD	Skewness	Kurtosis	JB_stat	JB_pvalue
r_HOSE	0.000371	0.012084	-0.979018	8.297453	2983.689910	0.000000
r_Gold	0.000417	0.031306	-0.830165	20.373266	28491.582157	0.000000
r_BTC	0.002526	0.047375	-0.718338	12.785279	9149.845877	0.000000

Table 3: Correlation matrix of return series

	r_HOSE	r_Gold	r_BTC
r_HOSE	1.00	-0.01	0.07
r_Gold	-0.01	1.00	-0.03
r_BTC	0.07	-0.03	1.00

Table 4: Unit root test results

Variable	Form	ADF stat.	ADF p-value	PP stat.	PP p-value	KPSS stat.	KPSS p-value
ln(VN-Index)	Level	-1.5407	0.5133	-1.6419	0.4613	5.8201	0.0100
	1st diff.	-13.4003	0.0000	-45.1399	0.0000	0.0837	0.1000
ln(Gold)	Level	1.3105	0.9967	-0.3202	0.9226	7.2542	0.0100
	1st diff.	-16.0764	0.0000	-85.8940	0.0000	0.3216	0.1000
ln(BTC/VND)	Level	-1.2077	0.6703	-1.2398	0.6563	6.5959	0.0100
	1st diff.	-49.8634	0.0000	-49.8402	0.0000	0.1335	0.1000

Table 4 presents the unit root test results using the ADF, PP, and KPSS frameworks. The three tests yield highly consistent results regarding the integration properties of the series. At log-levels (prices), all variables exhibit a unit root, as evidenced by the non-rejection of the null hypothesis in the ADF/PP tests and the rejection of stationarity in the KPSS test. However, upon taking the first difference (returns), all series become stationary at the 1% significance level, confirming that they are integrated of order one, $I(1)$. Because the price series share the same order of integration, utilizing a Vector Error Correction Model (VECM) instead of a traditional Vector Autoregressive (VAR) model is methodologically appropriate to capture both short-run dynamics and potential long-run cointegration without losing critical informational text through over-differencing.

4.2 Cointegration and VECM Results

4.2.1 Lag Length Selection

Prior to testing for cointegration, we select the lag order k of the VECM (10) directly, corresponding to $p - 1$ lagged differences in the error-correction representation. Appendix a1 reports the AIC, BIC, and Hannan–Quinn Information Criterion (HQIC) across VECM lag orders $k = 1, \dots, 10$. We rely on the HQIC as the primary selection criterion: with approximately $T = 2,000$ daily observations, the sample is sufficiently large that the HQIC’s asymptotic consistency dominates the AIC’s tendency to overfit, while its weaker penalty relative to the BIC better accommodates the rich short-run dynamics typical of daily financial data (Lütkepohl, 2005). The HQIC selects $k = 5$, and this lag order is maintained throughout all subsequent VECM estimations.

4.2.2 Johansen Cointegration Test

The Johansen trace test indicates one cointegrating relationship among the variables. Specifically, the null hypothesis of no cointegration ($r \leq 0$) is rejected at the 5% significance level because the trace statistic 30.7249 exceeds the 5% critical value 29.7961. However, the null hypothesis $r \leq 1$ cannot be rejected because the trace statistic 7.8407 is lower than the 5% critical value 15.4943. Therefore, the cointegration rank is selected as $r = 1$.

Table 5: Johansen Cointegration Trace Test Results

Null hypothesis	Trace statistic	5% critical value	Decision
$r \leq 0$	30.7249	29.7961	Reject H_0
$r \leq 1$	7.8407	15.4943	Do not reject H_0
$r \leq 2$	0.8181	3.8415	Do not reject H_0

4.2.3 Estimated Cointegrating Vector and Adjustment Coefficients

Table 6 presents the normalised cointegrating vector $\hat{\beta}$ and the adjustment coefficients $\hat{\alpha}$ from the estimated VECM (10), with the cointegrating equation normalised on $\ln P_t^{\text{HOSE}}$.

Table 6: Normalised Cointegrating Vector and Adjustment Coefficients

	Cointegrating vector $\hat{\beta}$	Adjustment $\hat{\alpha}$
$\ln P_t^{\text{HOSE}}$	1.0000 (normalised)	-0.0124**
$\ln P_t^{\text{BTC}}$	-0.1745	-0.0245*
$\ln P_t^{\text{Gold}}$	0.1891	-0.0069

Notes: Standard errors in parentheses. $\hat{\beta}$ is normalised on HOSE. $\hat{\alpha}$ measures the speed of adjustment toward long-run equilibrium; a negative and significant $\hat{\alpha}_i$ indicates that variable i is error-correcting. * $p < 0.05$; ** $p < 0.01$.

The estimated cointegrating equation takes the form:

$$\ln \hat{P}_t^{\text{HOSE}} = \hat{\beta}_0 - 0.1745 \ln P_t^{\text{BTC}} + 0.1891 \ln P_t^{\text{Gold}}, \quad (29)$$

Turning to the adjustment coefficients, $\hat{\alpha}_{\text{HOSE}}$ and $\hat{\alpha}_{\text{BTC}}$ is negative and statistically significant at the 5% level, while $\hat{\alpha}_{\text{Gold}}$ is insignificant.

4.2.4 VECM Diagnostic Tests

Table 7 summarises the diagnostic tests applied to the VECM residuals.

Table 7: VECM Residual Diagnostic Tests

Test	Null hypothesis	Statistic	p -value
LM autocorrelation (lag 1)	No serial correlation	0.5499	0.9999
LM autocorrelation (lag 5)	No serial correlation	14.9951	0.9999
ARCH-LM (lag 5) — HOSE	No ARCH effects	188.6321	7.66×10^{-39}
ARCH-LM (lag 5) — BTC	No ARCH effects	24.4009	0.0002
ARCH-LM (lag 5) — Gold	No ARCH effects	51.9610	5.50×10^{-10}
Jarque-Bera multivariate normality	Normality	3.794×10^4	< 0.001

The LM test does not reject the null of no serial correlation at either lag 1 or lag 5, confirming adequate lag specification. The ARCH-LM tests detect residual conditional heteroskedasticity in the BTC and HOSE equations, which is expected given the well-documented volatility clustering in daily financial returns (Engle,

1982); this does not invalidate the VECM estimates but motivates the use of HC standard errors in the safe-haven regression (Section 3.4). The multivariate Jarque–Bera test rejects normality, consistent with the excess kurtosis documented in Table 2; however, the Johansen estimator remains consistent under non-normality (Gonzalo, 1994), so inference on $\hat{\beta}$ and $\hat{\alpha}$ is unaffected. The companion matrix eigenvalue plot (Appendix Figure a1) confirms that all roots lie inside or on the unit circle, with exactly $r = 1$ unit root as required by the cointegration rank.

4.3 Long-run and Short-run Dynamics

4.3.1 Long-run Equilibrium Relationship

Section 4.3. The normalised cointegrating equation (29) reveals a well-defined long-run equilibrium linking the HOSE index, Bitcoin, and gold. Re-arranging to express the equilibrium error gives:

$$\hat{u}_t = \ln P_t^{\text{HOSE}} + 0.1745 \ln P_t^{\text{BTC}} - 0.1891 \ln P_t^{\text{Gold}} - \hat{\beta}_0, \quad (30)$$

so that a positive $\hat{u}_t$ indicates the HOSE index is above its long-run equilibrium level relative to BTC and gold.

The negative coefficient on $\ln P_t^{\text{BTC}}$ ($\hat{\beta}_1 = -0.1745$) implies a *positive* long-run co-movement between Bitcoin and the HOSE index: a permanent 1% increase in the Bitcoin price is associated with a 0.1745% increase in the long-run equilibrium level of the HOSE index. This positive relationship is consistent with growing evidence that Bitcoin increasingly co-moves with risk assets during periods of broad market expansion, reflecting the participation of retail investors who hold both equities and cryptocurrencies (Baur and McDermott, 2010).

By contrast, the positive coefficient on $\ln P_t^{\text{Gold}}$ ($\hat{\beta}_2 = 0.1891$) implies a *negative* long-run relationship between gold and the HOSE index after rearranging equation (30): a permanent rise in the gold price is associated with a lower long-run equilibrium for Vietnamese equities. This inverse relationship is consistent with gold’s traditional role as a counter-cyclical asset that appreciates during periods of economic uncertainty and risk-off sentiment, which simultaneously depress equity valuations (Baur and McDermott, 2010).

Adjustment dynamics and price discovery. The adjustment coefficients $\hat{\alpha}$ reveal a clear asymmetry in how the three assets respond to long-run disequilibria. The HOSE index carries the primary burden of error correction: $\hat{\alpha}_{\text{HOSE}} = -0.0124$ is negative and significant at the 1% level, implying that the Vietnamese stock market closes approximately 1.24% of any deviation from the long-run equilibrium each trading day, or equivalently, that roughly half of a shock is absorbed within $\ln(0.5)/\ln(1 - 0.0124) \approx 56$ trading days.

Bitcoin also exhibits statistically significant adjustment ($\hat{\alpha}_{\text{BTC}} = -0.0245$, $p < 0.05$), indicating that it is not a pure price-discovery leader in this system - it too responds, albeit more slowly in economic magnitude, to deviations from the trivariate equilibrium.

Gold, by contrast, has an insignificant adjustment coefficient ($\hat{\alpha}_{\text{Gold}} = -0.0069$), suggesting that gold prices are *weakly exogenous* with respect to the long-run relationship: gold is driven by global factors outside the Vietnamese market and does not adjust to correct local disequilibria. Weak exogeneity of gold is formally confirmed by a likelihood ratio test of the restriction $H_0: \alpha_{\text{Gold}} = 0$, following Johansen and Juselius (1992). This finding positions gold as the long-run *forcing variable* in the system, consistent with its status as a globally priced commodity whose price is set in international markets (Baur and McDermott, 2010).

4.3.2 Short-run Causal Dynamics

Table 8: Block Granger Causality Tests

Dependent variable	Excluded block	χ^2	p -value
$\Delta \ln P_t^{\text{HOSE}}$	Gold	1.128611	0.342618
	BTC	4.534805	0.000135**
	Gold + BTC	2.803098	0.000786**
$\Delta \ln P_t^{\text{Gold}}$	HOSE	0.628121	0.707931
	BTC	1.448395	0.191942
	HOSE + BTC	1.062333	0.387817
$\Delta \ln P_t^{\text{BTC}}$	HOSE	0.729099	0.626138
	Gold	1.908709	0.075536
	HOSE + Gold	1.311822	0.203673

Notes: The null hypothesis is that the excluded block does not Granger-cause the dependent variable.

* $p < 0.05$; ** $p < 0.01$.

The results of the Block Granger causality tests, presented in Table 8, reveal an asymmetric short-term transmission mechanism among the three assets. Specifically, Bitcoin exhibits a highly significant Granger causality towards the VN-Index (p -value < 0.01), indicating that historical price dynamics in the cryptocurrency market contain valuable predictive information for the Vietnamese stock market. Conversely, Gold does not Granger-cause the VN-Index, nor does it exhibit any significant reactions to lagged shocks from the other variables in the system. This empirical evidence suggests that Bitcoin plays a more active role in the short-term information transmission mechanism, whereas Gold operates largely independently, behaving predominantly as a weakly exogenous variable within the analyzed framework.

4.3.3 Generalised Impulse Response Functions

The 2.5th and 97.5th percentiles of the empirical distribution of $\widehat{\text{GIRF}}^{(b=1000)}(h=20)$ across bootstrap replications form the 95% confidence band reported in Figure 2. Although the impulse response values may

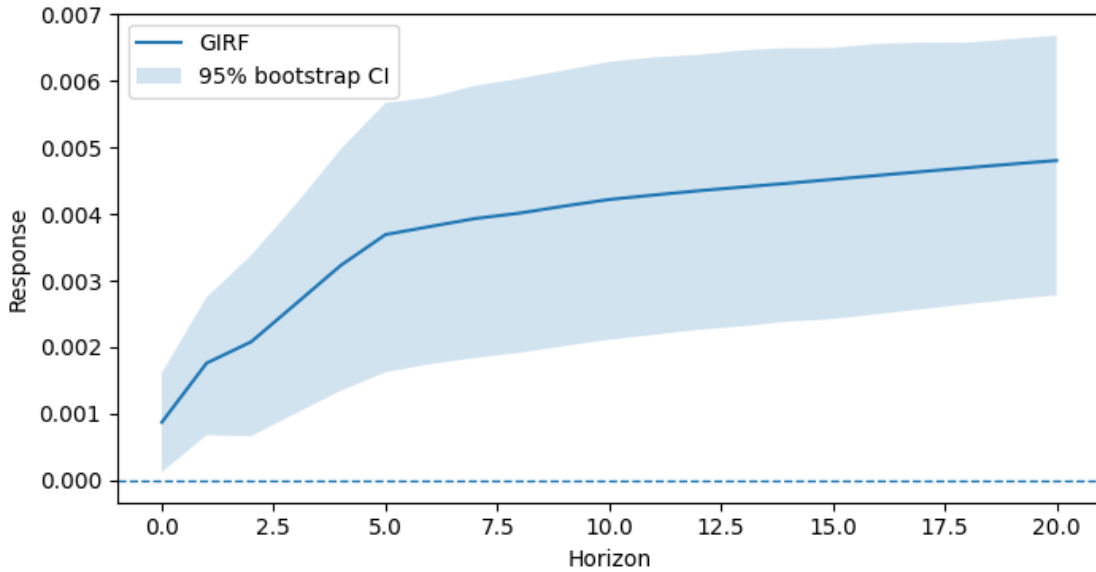


Figure 2: Generalised Impulse Response Functions

appear numerically marginal, it is important to note that all variables are log-transformed, allowing these

responses to be approximately interpreted as percentage changes. A shock to Bitcoin generates a positive response in the VN-Index, peaking at approximately 0.5% after 20 periods. The bootstrap confidence interval for this trajectory remains strictly above zero, indicating a statistically significant effect. Conversely, the response of the VN-Index to a Gold shock, as well as the reciprocal responses of both Bitcoin and Gold to a VN-Index shock, lack statistical significance, given that their respective confidence intervals encompass zero. These dynamic interactions are visually corroborated by the impulse response charts presented in the Appendix.

4.3.4 Generalised Forecast Error Variance Decomposition

The Generalized Forecast Error Variance Decomposition (GFEVD) results indicate that the forecast error variance of each variable is predominantly driven by its own innovations. Nevertheless, regarding cross-market spillovers, Bitcoin contributes the most notable share to the forecast variance of the HOSE. Specifically, at a 10-period horizon, shocks from Bitcoin account for 6.15% of the forecast error variance of the VN-Index, a proportion that escalates to 9.63% at the 20-period horizon. In stark contrast, innovations originating from Gold explain a mere 0.40% and 0.72% over the respective horizons. These findings suggest that Bitcoin exerts a more pronounced influence than Gold in transmitting short-term shocks to the Vietnamese equity market, although the magnitude of this transmission mechanism remains relatively modest.

Table 9: Generalised Forecast Error Variance Decomposition

Horizon	Variance of	Shock from HOSE	Shock from Gold	Shock from BTC
$H = 10$	HOSE	93.45%	0.40%	6.15%
	Gold	0.12%	99.69%	0.19%
	BTC	0.41%	0.12%	99.47%
$H = 20$	HOSE	89.65%	0.72%	9.63%
	Gold	0.08%	99.69%	0.23%
	BTC	0.22%	0.13%	99.65%

Notes: Entries are normalised GFEVD shares $\hat{\theta}_{ij}^g(10)$ from equation (13). Rows need not sum to 100% due to the generalised (non-orthogonal) decomposition (Pesaran and Shin, 1998).

4.4 Safe-haven Stress Test Results

Table 10 reports the estimated total slopes $\hat{\beta}_0 + \hat{\beta}_q$ from equation (14) for gold and Bitcoin, together with heteroskedasticity-consistent standard errors (White, 1980). A negative and statistically significant total slope at quantile q indicates that the asset qualifies as at least a *weak safe haven* for the HOSE index during stress at that threshold (Baur and McDermott, 2010).

4.4.1 Gold as a Safe Haven

The results provide partial support for gold’s safe-haven role in the Vietnamese stock market. Under normal market conditions, gold’s co-movement with the HOSE index is positive but statistically insignificant ($\hat{\beta}_0 = 0.0440$, $p = 0.621$), indicating no systematic directional relationship in tranquil periods.

Crucially, the total slope turns negative and statistically significant at the 10% stress threshold ($\hat{\beta}_0 + \hat{\beta}_1 = -0.2569$, $p = 0.030$), satisfying the Baur and McDermott (2010) criterion for a *weak safe haven*: during moderate market stress, gold’s return moves in the opposite direction to the HOSE index, providing a degree of portfolio protection precisely when it is needed.

However, this safe-haven property does not persist at more extreme stress levels. At the 5% and 1% stress thresholds, the total slopes remain negative (-0.1213 and -0.1811 , respectively) but lose statistical significance ($p = 0.234$ and $p = 0.343$). The attenuation of the safe-haven signal at the 1% quantile may reflect the dynamics of severe global crises — such as the COVID-19 market crash of March 2020 — during which correlations across virtually all asset classes temporarily converged toward unity as investors liquidated positions indiscriminately to meet margin calls (Akhtaruzzaman et al., 2021). Overall, gold qualifies as a

weak safe haven for Vietnamese equities under moderate stress, but its protective capacity is not statistically robust during the most extreme market dislocations.

4.4.2 Bitcoin as a Safe Haven

Bitcoin, by contrast, exhibits no safe-haven characteristics for the HOSE index at any stress threshold. The normal-period slope is positive and insignificant ($\hat{\beta}_0 = 0.1655$, $p = 0.214$), consistent with the absence of a stable unconditional relationship identified in the VECM short-run dynamics (Section 4.3.2).

More strikingly, rather than declining during market stress, Bitcoin’s total co-movement slope *increases* at the 10% threshold ($\hat{\beta}_0 + \hat{\beta}_1 = 0.2842$) and rises sharply at the 1% extreme-stress threshold ($\hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3 = 0.9096$, $p = 0.072$). Although the 1% slope falls just short of conventional significance, its magnitude — nearly one-to-one co-movement with the HOSE index at the most severe stress quantile — is economically large and directionally unambiguous. This pattern is consistent with the “contagion” hypothesis: during extreme market episodes, Bitcoin behaves as a *risk-on* asset that amplifies rather than absorbs equity market losses (Baur and Lucey, 2010; Akhtaruzzaman et al., 2021).

Table 10: Safe-haven Stress Test: Total Slopes at Stress Thresholds

Asset	Slope	Estimate	Std. Err.	<i>t</i> -stat	<i>p</i> -value
Gold	Normal ($\hat{\beta}_0$)	0.0440	0.0890	0.4946	0.6209
	10% stress ($\hat{\beta}_0 + \hat{\beta}_1$)	−0.2569	0.1183	−2.1721	0.0298*
	5% stress ($\hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2$)	−0.1213	0.1020	−1.1896	0.2342
	1% stress ($\hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3$)	−0.1811	0.1909	−0.9487	0.3428
Bitcoin	Normal ($\hat{\beta}_0$)	0.1655	0.1331	1.2434	0.2137
	10% stress ($\hat{\beta}_0 + \hat{\beta}_1$)	0.2842	0.2641	1.0759	0.2820
	5% stress ($\hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2$)	0.0145	0.2079	0.0699	0.9443
	1% stress ($\hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3$)	0.9096	0.5056	1.7990	0.0720

Notes: * $p < 0.05$.

4.4.3 Comparative Assessment

The divergent findings for gold and Bitcoin carry clear implications for Vietnamese investors seeking to hedge equity market risk. Gold satisfies the weak safe-haven criterion of Baur and Lucey (2010) at the 10% stress threshold, making it a viable-if partial-hedging instrument for moderate downturns. Bitcoin, however, fails to serve this function and may actively worsen portfolio drawdowns during extreme stress events.

4.5 Dynamic Correlation Results

Figure 3 presents the time-varying EWMA (RiskMetrics) correlations between the HOSE index and the two candidate safe-haven assets, namely gold and Bitcoin. The dynamic correlation patterns provide important insights into how these assets behave during periods of market stress.

A particularly informative episode is the outbreak of the COVID-19 pandemic in March 2020. During this period, the correlation between Bitcoin and the HOSE index surged sharply, reaching its highest level over the sample period, with values approaching 0.7. This indicates that Bitcoin moved closely with the Vietnamese equity market during the crisis, suggesting that investors liquidated both risky assets simultaneously in response to heightened uncertainty and liquidity needs. In contrast, the HOSE–Gold correlation declined dramatically and became strongly negative, reaching approximately −0.6. This divergence suggests a flight-to-safety effect, where capital exited equities and flowed into gold, reinforcing gold’s traditional role as a defensive asset during severe market turmoil.

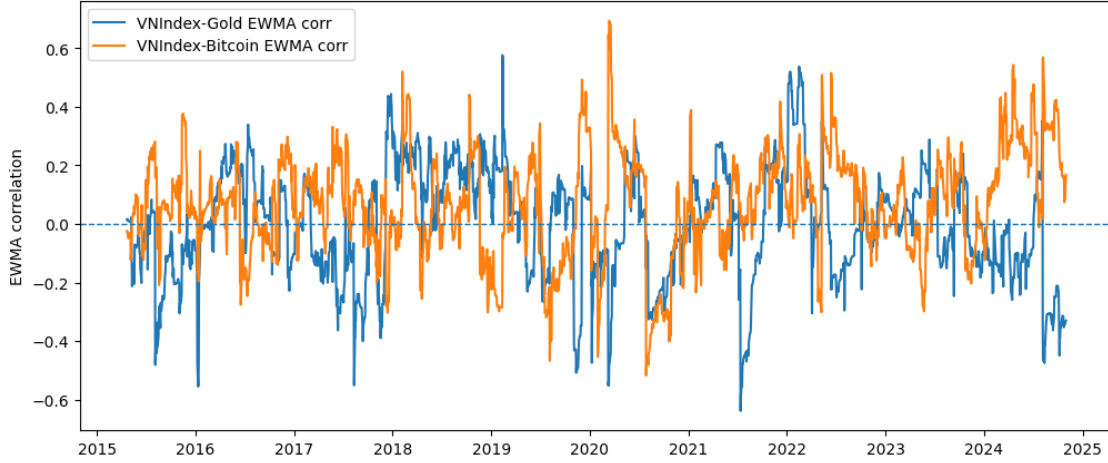


Figure 3: EWMA correlation series

A similar pattern emerged during the 2022 market downturn, which coincided with the global cryptocurrency collapse and tightening monetary conditions. Throughout this period, the HOSE–Bitcoin correlation remained persistently positive and frequently fluctuated between 0.4 and 0.6, indicating that Bitcoin behaved more like a speculative risk asset than a safe haven. Meanwhile, the equity–gold correlation experienced another pronounced decline and again approached -0.6 toward the end of 2022, suggesting renewed demand for gold as a portfolio protection instrument during episodes of extreme market stress.

Table 11 further supports this interpretation. During normal conditions, the average correlation between gold and the domestic stock market is close to zero, suggesting relative independence between the two assets. Under the 10% and 5% stress regimes, the correlations become slightly positive, implying that the safe-haven property is not consistently observed across moderate downturns. However, under the most extreme stress regime (1%), the correlation turns negative, with a mean value of approximately -0.052 and a median value of approximately -0.056 . This suggests that gold may provide stronger downside protection during the most severe episodes of market panic.

By contrast, Bitcoin exhibits no clear safe-haven characteristics. The average Bitcoin–HOSE correlation remains positive across all market conditions and increases with market stress, rising from approximately 0.065 during normal periods to around 0.126 under the most extreme stress regime. Taken together, these findings suggest that Bitcoin behaves predominantly as a risky asset rather than a safe haven for Vietnamese equity investors.

Overall, the combined evidence from the safe-haven regressions and the EWMA correlation analysis suggests that gold exhibits stronger safe-haven properties than Bitcoin. However, the evidence is not sufficiently robust to conclude that gold acts as an absolute safe haven across all market conditions. Gold appears to provide a more effective hedging role under certain stress regimes, particularly under the 10% stress threshold in the regression-based framework and under the 1% extreme stress regime in the EWMA correlation analysis. By contrast, Bitcoin does not display clear safe-haven characteristics. Instead, Bitcoin tends to move positively with the HOSE index during periods of market stress, implying that it may behave more like a risk-sensitive asset. These findings suggest that Bitcoin may contribute to short-term risk transmission to the Vietnamese stock market, but it cannot be regarded as a reliable safe-haven alternative to gold.

Table 11: EWMA Correlation across Market Stress Regimes

Regime	N	HOSE–Gold			HOSE–Bitcoin			Gold–Bitcoin
		Mean	Median	Negative Share	Mean	Median	Negative Share	Mean
Normal	1957	-0.0067	-0.0012	0.5023	0.0648	0.0753	0.3316	-0.0264
Stress 10%	223	0.0186	0.0380	0.4439	0.0725	0.0498	0.3453	-0.0037
Stress 5%	112	0.0195	0.0435	0.4464	0.0798	0.0483	0.3661	0.0086
Stress 1%	23	-0.0516	-0.0558	0.5652	0.1257	0.0273	0.4348	-0.0091

4.6 Extension Results

4.6.1 Rolling VECM: Time-varying Adjustment and Causality

Figure 4 plots the time-varying adjustment coefficients $\hat{\alpha}_{i,t}$ extracted from the rolling VECM estimation, and Figure 5 traces the rolling Granger causality p -values for the Bitcoin $\rightarrow$ VN-Index and Gold $\rightarrow$ VN-Index channels. Together, these figures reveal that the long-run and short-run relationships identified in the full-sample VECM (Sections 4.3–4.3.2) mask substantial time variation.

Time-varying adjustment coefficients. The rolling $\hat{\alpha}$ series in Figure 4 fluctuate considerably across the sample period. The VN-Index adjustment coefficient $\hat{\alpha}_{\text{HOSE},t}$ remains close to zero and relatively stable throughout most of the sample, consistent with its modest full-sample estimate of -0.0124 . Gold’s rolling $\hat{\alpha}_{\text{Gold},t}$, by contrast, exhibits pronounced spikes in both directions, with the most dramatic episode occurring in early 2020, when the coefficient plunges to approximately -0.22 before rapidly reverting-reflecting a transitory but sharp realignment of gold prices relative to the trivariate equilibrium during the COVID-19 market dislocation. Bitcoin’s rolling $\hat{\alpha}_{\text{BTC},t}$ similarly shows heightened variability around the COVID-19 period and again during the 2022 cryptocurrency market correction, suggesting that the cointegrating relationship intermittently breaks down and re-establishes itself following major market events.

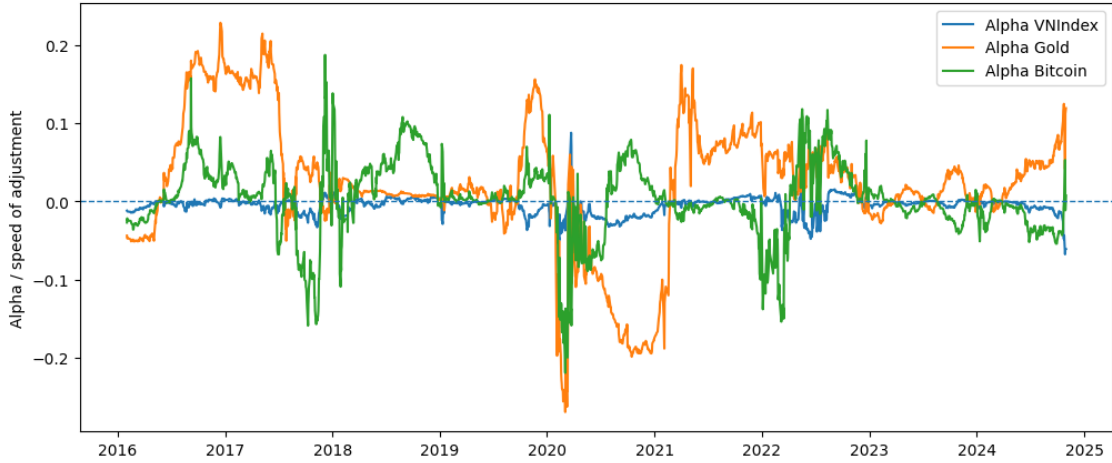


Figure 4: Rolling speed of adjustment

Time-varying Granger causality. Figure 5 plots the rolling p -values of the Granger causality test for the Bitcoin $\rightarrow$ VN-Index and Gold $\rightarrow$ VN-Index channels, with the horizontal dashed line marking the 5% significance threshold. The results reveal marked time variation in both channels.

For the Bitcoin $\rightarrow$ VN-Index channel, the p -value remains above 0.05 for much of the pre-COVID period, indicating that Bitcoin’s short-run predictive power for Vietnamese equity returns is episodic rather than continuous. A decisive shift occurs around the onset of the COVID-19 pandemic in early 2020: the p -value drops sharply and remains almost continuously below the 5% threshold throughout 2020–2021, with a rolling window significance rate of 99.78% and a mean p -value of just 0.0043 during this sub-period. This implies that Bitcoin served as a near-perfect short-run leading indicator for VN-Index returns at the height of the

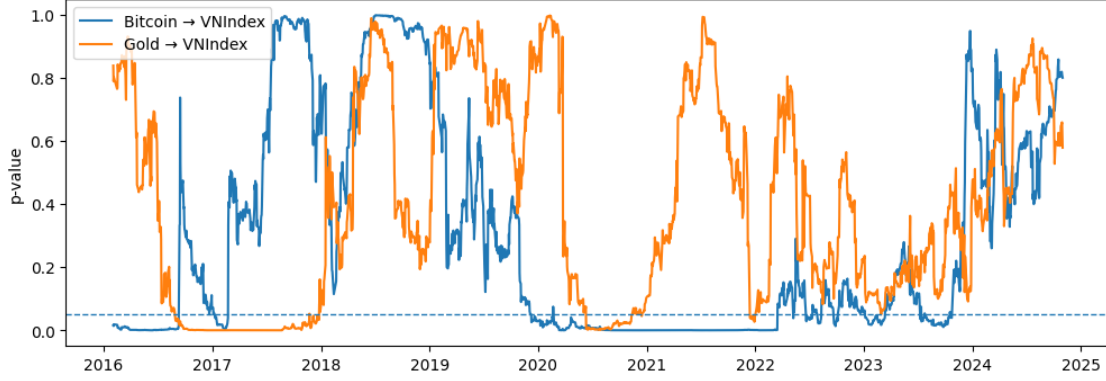


Figure 5: Rolling $P_{value of Granger}$

pandemic—a period of exceptional global uncertainty when cryptocurrency markets, operating around the clock, transmitted risk signals ahead of the next opening of the Vietnamese stock exchange (Akhtaruzzaman et al., 2021). The channel weakens markedly after 2021, reappearing intermittently in 2022–2023 before fading again in 2024, confirming that Bitcoin’s informational role is crisis-contingent rather than structurally embedded in the Vietnamese market. Across the full sample, Bitcoin Granger-causes VN-Index returns in approximately 40.35% of rolling windows, compared with 21.50% for gold.

For the Gold → VN-Index channel, the p -value trajectory in Figure 5 shows a broadly similar crisis-contingent pattern but with lower overall frequency of significance (21.50% of windows), corroborating the view that gold’s connection to the Vietnamese equity market is weaker and more sporadic than Bitcoin’s during stress episodes.

4.6.2 Diebold–Yilmaz Spillover Analysis

Table 12 reports the directional spillover summary from the Diebold–Yilmaz framework (Diebold and Yilmaz, 2012b, 2014) at forecast horizons $H = 10$ and $H = 20$ trading days, estimated using the VECM-implied moving-average representation (equation (13)).

Overall connectedness. The Total Connectedness Index (TCI) is low at both horizons — 2.46% at $H = 10$ and 3.67% at $H = 20$ — indicating that the vast majority of each asset’s forecast error variance is driven by its own idiosyncratic shocks rather than cross-market spillovers. The modest increase from $H = 10$ to $H = 20$ suggests that cross-market shock transmission, while limited, is somewhat more pronounced at longer horizons as information diffuses more completely across markets. The low overall TCI is consistent with the segmented nature of the Vietnamese equity market (Bekaert et al., 2002) and the relatively recent integration of Bitcoin into Vietnamese retail portfolios (Corbet et al., 2018).

Bitcoin as the dominant transmitter. Despite the low aggregate connectedness, Bitcoin emerges as the dominant source of cross-market shock transmission in the system. At $H = 10$, Bitcoin transmits 6.33 percentage points to other variables while receiving only 0.53, yielding a net spillover of +5.80 — by far the largest in the system. This asymmetry intensifies at $H = 20$, where Bitcoin’s net spillover rises to +9.51, driven primarily by the increasing contribution of Bitcoin shocks to VN-Index forecast error variance (6.15% at $H = 10$, rising to 9.63% at $H = 20$). The growing net transmission at longer horizons implies that Bitcoin’s influence on the Vietnamese stock market materialises gradually, consistent with the delayed information processing typical of less liquid emerging equity markets.

VN-Index as the dominant receiver. The VN-Index is a clear net receiver at both horizons, absorbing −6.01 net percentage points at $H = 10$ and −10.05 at $H = 20$. The worsening receiver position at $H = 20$ mirrors Bitcoin’s expanding transmitter role, confirming a directional and asymmetric relationship: shocks originate predominantly in the Bitcoin market and are subsequently absorbed by the Vietnamese equity market.

Gold’s limited transmission role. Gold is a net transmitter in both panels, but its net spillover is negligible relative to Bitcoin (+0.21 at $H = 10$, +0.54 at $H = 20$). Gold transmits to and receives from other assets at similarly low levels, consistent with its weak exogeneity identified in the VECM (Section 4.3.1) and its safe-haven characterisation in Section 4.4: gold’s relative independence from the system means it neither meaningfully amplifies nor absorbs short-run shocks between Bitcoin and Vietnamese equities.

Table 12: Diebold–Yilmaz Directional Spillover Summary

Variable	Horizon H			
	TO others	FROM others	NET	Role
<i>Panel A: $H = 10$ trading days</i>				
HOSE	0.53	6.55	−6.01	Net receiver
Gold	0.52	0.31	+0.21	Net transmitter
Bitcoin	6.33	0.53	+5.80	Net transmitter
Total Connectedness Index (TCI): 2.46%				
<i>Panel B: $H = 20$ trading days</i>				
HOSE	0.30	10.35	−10.05	Net receiver
Gold	0.85	0.31	+0.54	Net transmitter
Bitcoin	9.86	0.35	+9.51	Net transmitter
Total Connectedness Index (TCI): 3.67%				

Notes: Entries are percentage-point contributions to forecast error variance decomposition, based on normalised GFEVD $\tilde{\theta}_{ij}^g(H)$ from equation (25)–(28). NET = TO − FROM. A positive NET indicates a net transmitter of shocks; negative NET indicates a net receiver.

Synthesis. Taken together, the rolling VECM and Diebold–Yilmaz results reinforce and extend the findings from the full-sample analysis. Bitcoin plays a dual role in the Vietnamese financial system: a crisis-contingent Granger-causal predictor of equity returns (concentrated in the COVID-19 period) and a persistent net transmitter of forecast error variance across all horizons. Gold, by contrast, operates largely in isolation—a weak safe-haven asset (Section 4.4) that remains disconnected from the short-run dynamics of both Bitcoin and the Vietnamese equity market.

4.7 Robustness Checks

To ensure the validity, stability, and reliability of the main empirical findings, a comprehensive set of robustness checks is conducted. The detailed estimation outputs for all alternative specifications are reported in the Appendix.

Potential concerns regarding model specification and currency effects are addressed through alternative macro-financial setups. A standard Vector Autoregression (VAR) model is estimated using raw returns to verify whether the short-run dynamic conclusions are driven by the baseline VECM specification. In addition, the BTC/VND series is replaced by the global BTC/USD price, while the USD/VND exchange rate is included as an exogenous control variable. This specification helps rule out the possibility that the observed co-movements are mainly driven by domestic currency fluctuations. The results obtained from these alternative setups remain highly consistent with the baseline findings.

The sensitivity of the estimated results is further examined by varying key model parameters. For the dynamic correlation analysis, the Exponentially Weighted Moving Average (EWMA) correlations are re-estimated using alternative decay factors of $\lambda = 0.90$ and $\lambda = 0.97$, instead of the baseline RiskMetrics standard of $\lambda = 0.94$. The statistical significance of the safe-haven stress tests is also reassessed using Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors with alternative lag lengths of 10 and 20 days. Across these parameter variations, the core conclusions remain quantitatively and qualitatively stable: gold exhibits stronger safe-haven properties, whereas Bitcoin behaves predominantly as a risk-sensitive asset.

The effect of temporal aggregation is also evaluated by re-estimating the VAR framework using weekly data. Although the overall results are broadly consistent with the daily baseline, the weekly VAR reveals

a more nuanced dynamic. In particular, the volatility spillover from Bitcoin to the HOSE index becomes notably weaker at the lower frequency. This finding suggests that shocks from the cryptocurrency market may transmit rapidly to domestic equities in the short run, but their contagion effects appear to be short-lived and are largely absorbed within a few trading days. Therefore, the evidence does not support a persistent long-term structural impact from Bitcoin shocks on the Vietnamese equity market.

5 Conclusion

This study contributes to the “Digital Gold” debate by examining the safe-haven properties and dynamic spillovers among Bitcoin, gold, and the Vietnamese stock market, represented by the VN-Index. Using a combination of VECM, Granger causality tests, EWMA dynamic correlations, and the Diebold-Yilmaz spillover framework, the findings provide evidence that Bitcoin may not function as a close substitute for gold in the context of portfolio risk management in Vietnam.

The VECM results suggest the existence of a long-run equilibrium relationship among Bitcoin, gold, and the VN-Index. Within this long-run structure, gold appears to maintain a negative association with the Vietnamese stock market, which is broadly consistent with its traditional role as a counter-cyclical asset and a potential hedge during periods of market weakness. By contrast, Bitcoin shows a positive long-run relationship with the VN-Index, implying that it may behave more like a risk asset than a defensive instrument in the Vietnamese market. In addition, gold seems to act relatively independently from short-run market adjustments, while Bitcoin exhibits some predictive power for movements in the VN-Index, particularly during periods of heightened uncertainty such as the COVID-19 crisis.

The spillover analysis further supports this distinction between Bitcoin and gold. Although the overall level of connectedness among the three assets is relatively limited, the Diebold-Yilmaz results indicate that Bitcoin tends to act as a net transmitter of shocks to the Vietnamese stock market. The VN-Index, in turn, appears to receive volatility shocks from Bitcoin, whereas gold plays only a minor role in short-run risk transmission. These results suggest that Bitcoin may contribute to the propagation of financial stress rather than insulating domestic equity investors from it.

Regarding safe-haven properties, the stress-test results and EWMA dynamic correlations show a noticeable contrast between gold and Bitcoin. Gold provides some evidence of a weak safe-haven role for the VN-Index, particularly during moderate market downturns. During crisis episodes such as the COVID-19 shock and the 2022 market decline, the behavior of gold is consistent with a flight-to-safety mechanism, in which investors may shift from equities toward more traditional defensive assets. Bitcoin, however, does not display comparable safe-haven characteristics. In more severe market stress conditions, Bitcoin tends to move in the same direction as the VN-Index, suggesting that it may amplify portfolio losses rather than reduce them.

Overall, the findings imply that the label “Digital Gold” should be applied to Bitcoin with caution, at least from the perspective of Vietnamese equity investors. While gold continues to offer diversification and partial protection during periods of market turbulence, Bitcoin appears to be more closely associated with speculative risk and short-run shock transmission. Therefore, investors should be careful when treating Bitcoin as a safe-haven asset in portfolio allocation decisions. Similarly, policymakers and risk managers may need to consider the potential role of cryptocurrency markets in transmitting financial volatility to domestic asset markets.

6 Limitations and Future Research

Although this study provides useful evidence on the safe-haven properties and dynamic spillovers among Bitcoin, gold, and the VN-Index, several limitations should be acknowledged. First, the analysis is limited to three assets, namely Bitcoin, gold, and the Vietnamese stock market. This focused setting allows the study to directly address the “Digital Gold” debate, but it may not fully capture broader portfolio substitution effects involving other defensive or risky assets, such as government bonds, the U.S. dollar, oil, commodity indices, stablecoins, or other major cryptocurrencies. Previous studies suggest that the role of Bitcoin as a portfolio asset can depend strongly on the asset universe, portfolio constraints, and the modelling framework used (Henriques and Sadorsky, 2018; Wang et al., 2022). Therefore, the findings of this study should be

interpreted mainly from the perspective of Vietnamese equity investors rather than as a general conclusion for all investment portfolios.

Second, this research relies primarily on historical price dynamics and does not explicitly incorporate market frictions such as transaction costs, liquidity constraints, taxation, bid-ask spreads, and regulatory uncertainty. These factors may be particularly relevant in Vietnam, where the practical accessibility and legal treatment of cryptocurrency investment remain more complex than those of traditional assets such as gold. While portfolio-based studies have shown that the economic value of Bitcoin should be evaluated after considering trading costs and portfolio turnover (Henriques and Sadorsky, 2018), this study mainly focuses on statistical relationships and therefore does not directly measure the net economic benefit of holding Bitcoin or gold in a real investment portfolio.

Third, the econometric framework used in this study may not fully capture nonlinear dependence and tail-risk behaviour. The VECM is suitable for examining long-run cointegration and short-run adjustment among non-stationary variables, but it remains largely linear in nature. Similarly, the EWMA rolling correlation provides a practical measure of time-varying dependence, yet it may not fully reflect asymmetric co-movements under extreme market conditions. This limitation is important because safe-haven behaviour is usually most relevant during market stress, when dependence may differ across quantiles and investment horizons. Recent studies have therefore employed quantile-based spillover models and wavelet quantile correlation to capture tail dependence and frequency-specific safe-haven behaviour (Urom et al., 2020; Kumar and Padakandla, 2022).

Fourth, although the Diebold–Yilmaz spillover index helps identify whether Bitcoin, gold, or the VN-Index acts as a net transmitter or receiver of shocks, its results may still be sensitive to lag selection, forecast horizon, volatility proxy, and sample period. In addition, the use of a VECM-based generalized forecast error variance decomposition provides consistency with the cointegration framework, but it may not fully capture structural breaks or regime shifts. This is relevant because previous research shows that spillover effects are time-varying and tend to intensify during extreme events such as the European debt crisis and the COVID-19 pandemic (Wang et al., 2022; Urom et al., 2020).

Future research could extend this study in several directions. First, future studies may apply quantile spillover models, cross-quantilogram methods, or wavelet quantile correlation to examine whether the safe-haven role of Bitcoin and gold changes across normal, bearish, and bullish market states. This would allow future research to distinguish average-market dependence from lower-tail dependence, which is more directly related to safe-haven behaviour (Urom et al., 2020; Kumar and Padakandla, 2022). Second, frequency-domain methods could be used to compare short-run panic reactions with longer-term diversification effects, since investors with different horizons may respond differently to market stress.

Third, future research could employ time-varying parameter VAR, regime-switching models, or nonlinear connectedness frameworks to better capture structural breaks and crisis-specific dynamics. This would be particularly useful for examining whether the relationships among Bitcoin, gold, and Vietnamese equities changed during COVID-19, the 2022 crypto-market correction, or domestic stock-market downturns. Since TVP-VAR models can capture time-varying spillovers without relying on a fixed rolling-window size, they may provide a more flexible framework for identifying changing transmitter–receiver roles across financial markets (Wang et al., 2022).

Fourth, the asset universe could be broadened by including Ethereum, stablecoins, commodity indices, exchange rates, interest rates, global uncertainty indices, and regional equity markets. Such an extension would help determine whether Bitcoin’s limited safe-haven role is specific to the VN-Index or reflects a broader pattern across emerging markets. Finally, future studies could move beyond statistical connectedness by testing the economic value of Bitcoin and gold in actual portfolio allocation. This may include out-of-sample forecasting, minimum-variance portfolio construction, downside-risk measures, performance-fee analysis, and transaction-cost adjustments. These extensions would provide a more comprehensive assessment of whether Bitcoin can play a meaningful role in risk management, rather than merely displaying short-term statistical relationships with the Vietnamese stock market.

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7 Appendix

7.1 Bonus result

Table a1: VECM Lag Order Selection

Lag Order	AIC	BIC	FPE	HQIC
0	-21.90	-21.86*	3.095×10^{-10}	-21.88
1	-21.91	-21.85	3.047×10^{-10}	-21.89
2	-21.91	-21.83	3.046×10^{-10}	-21.88
3	-21.93	-21.82	2.999×10^{-10}	-21.89
4	-21.96	-21.83	2.907×10^{-10}	-21.91
5	-21.97	-21.82	2.866×10^{-10}	-21.92*
6	-21.98	-21.80	2.858×10^{-10}	-21.91
7	-21.98	-21.78	2.843×10^{-10}	-21.91
8	-21.98	-21.76	2.835×10^{-10}	-21.90
9	-21.99*	-21.74	$2.831^* \times 10^{-10}$	-21.90
10	-21.98	-21.71	2.838×10^{-10}	-21.88

Notes: The table reports the lag order selection results for the VECM specification. AIC, BIC, FPE, and HQIC denote the Akaike Information Criterion, Bayesian Information Criterion, Final Prediction Error, and Hannan–Quinn Information Criterion, respectively. The asterisk indicates the minimum value for each criterion.

Table a2: LM Test for Residual Autocorrelation

Lag Order	LM Statistic	df	<i>p</i> -value
1	0.549869	9	0.999954
2	2.540529	18	0.999992
3	6.277811	27	0.999988
4	10.263605	36	0.999992
5	14.995146	45	0.999993

Notes: The table reports the LM test results for system residual autocorrelation. The null hypothesis is H_0 : no system residual autocorrelation. High p -values indicate that the null hypothesis cannot be rejected, suggesting no evidence of residual autocorrelation in the system.

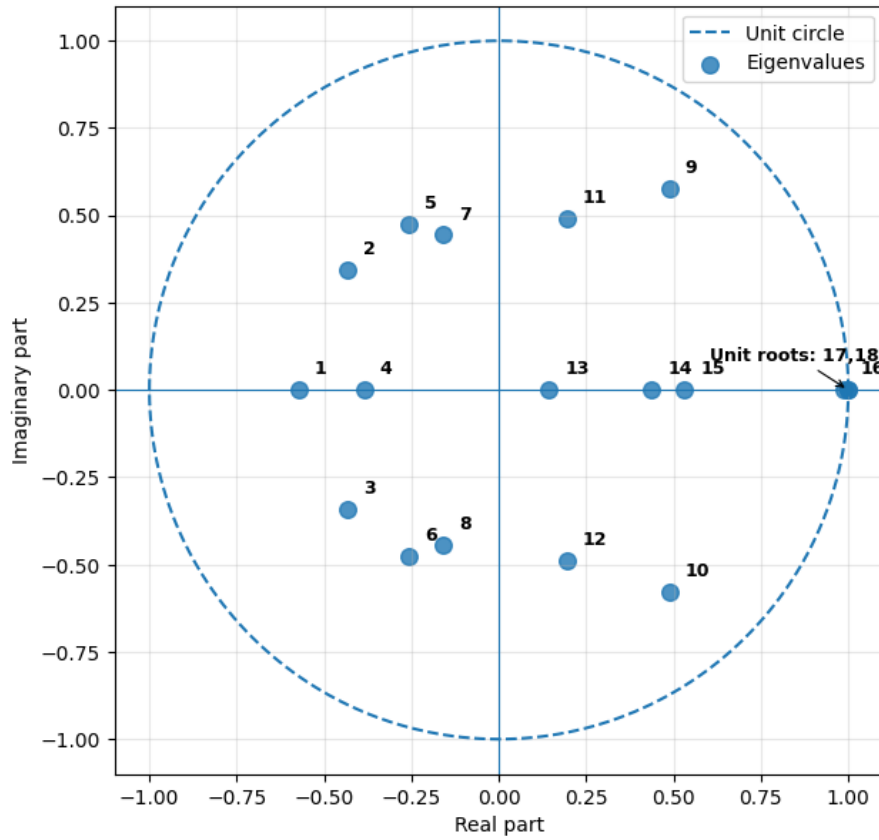


Figure a1: Eigenvalues of the Companion Matrix from the VECM-Implied VAR Representation

Figure a1 plots the eigenvalues of the companion matrix associated with the VECM-implied VAR representation. The dashed circle denotes the unit circle. Most eigenvalues lie inside the unit circle, indicating that the short-run dynamic components of the system are stable.

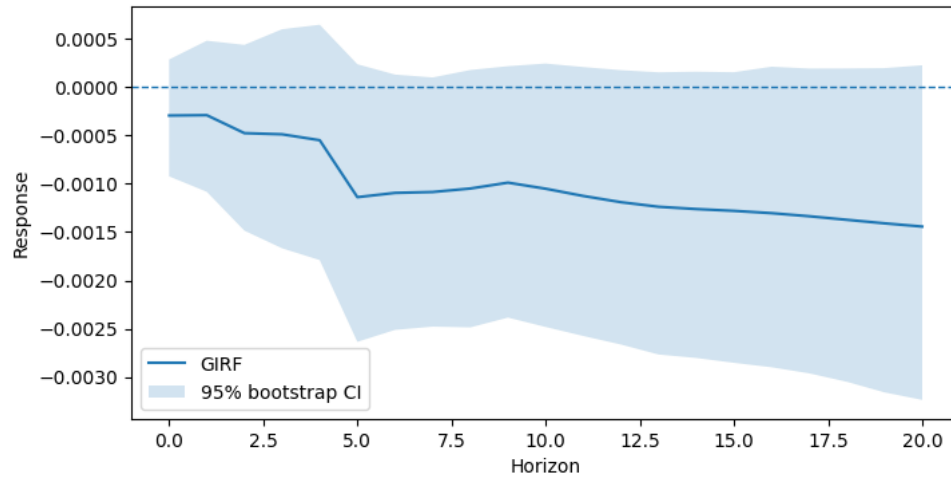


Figure a2: Generalised Impulse Response of the VN-Index to a Gold Shock

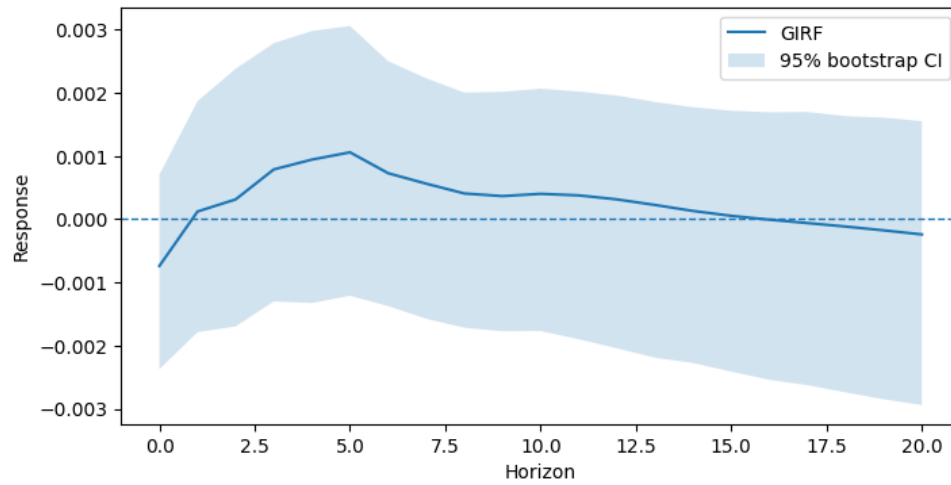


Figure a3: Generalised Impulse Response of the Gold to a VN-Index Shock

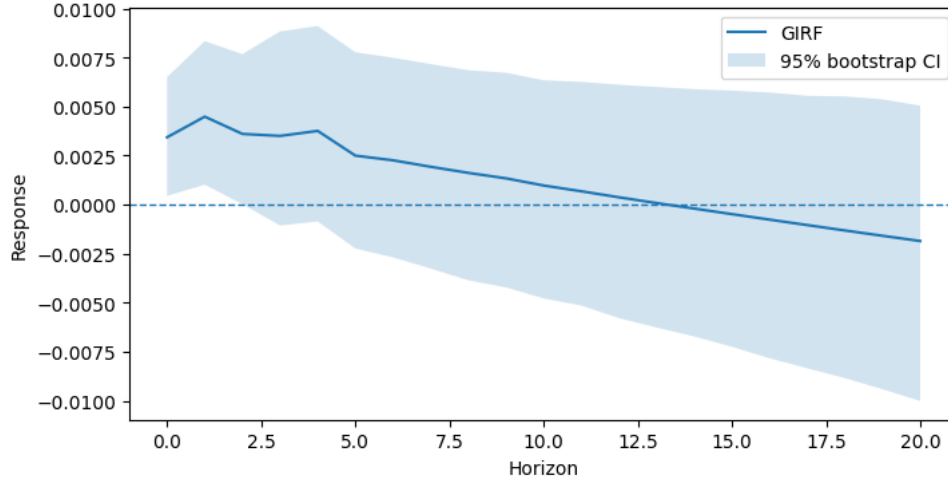


Figure a4: Generalised Impulse Response of the BTC to a VN-Index Shock

Table a3: Period-Specific Significance of Directional Predictability toward the HOSE Index

Period	N	Bitcoin → HOSE		Gold → HOSE	
		Mean p -Value	Significant Share	Mean p -Value	Significant Share
COVID	460	0.0043	0.9978	0.3939	0.2587
Post-COVID	628	0.2487	0.2691	0.3767	0.0000
Pre-COVID	907	0.4775	0.1951	0.4137	0.3418

Notes: The table reports the period-specific average p -values and the share of statistically significant tests for directional predictability toward the HOSE index. Bitcoin → HOSE and Gold → HOSE denote the predictive directions from Bitcoin and gold to the Vietnamese equity market, respectively. Significant Share refers to the proportion of rolling-window tests in which the null hypothesis of no directional predictability is rejected at the 5% significance level.

Table a4: Period-Specific Adjustment Coefficients in the Rolling VECM

Period	HOSE		Gold		Bitcoin	
	Mean α	Significant Share	Mean α	Significant Share	Mean α	Significant Share
COVID	-0.0090	0.2652	-0.0362	0.9674	-0.0130	0.0652
Post-COVID	-0.0048	0.1242	0.0261	0.8901	-0.0026	0.5032
Pre-COVID	-0.0036	0.1676	0.0517	0.9890	0.0159	0.1566

Notes: The table reports the period-specific average adjustment coefficients from the rolling VECM estimation. The coefficient α measures the speed at which each variable adjusts to deviations from the long-run cointegrating equilibrium. Significant Share denotes the proportion of rolling-window estimates in which the corresponding adjustment coefficient is statistically significant at the 5% level.

7.2 Robustness Check Results

Table a5: VAR Lag Order Selection

Lag Order	AIC	BIC	FPE	HQIC
0	-21.88	-21.88*	3.136×10^{-10}	-21.88
1	-21.90	-21.87	3.079×10^{-10}	-21.89
2	-21.90	-21.85	3.073×10^{-10}	-21.88
3	-21.92	-21.84	3.019×10^{-10}	-21.89
4	-21.95	-21.86	2.919×10^{-10}	-21.92
5	-21.97	-21.85	2.875×10^{-10}	-21.92*
6	-21.97	-21.83	2.867×10^{-10}	-21.92
7	-21.98	-21.81	2.852×10^{-10}	-21.92
8	-21.98	-21.79	2.847×10^{-10}	-21.91
9	-21.98*	-21.77	$2.840^* \times 10^{-10}$	-21.90
10	-21.98	-21.74	2.847×10^{-10}	-21.89

Notes: The table reports the lag order selection results for the VAR specification. AIC, BIC, FPE, and HQIC denote the Akaike Information Criterion, Bayesian Information Criterion, Final Prediction Error, and Hannan–Quinn Information Criterion, respectively. The asterisk indicates the minimum value for each criterion.

Table a6: VAR-Based Granger Causality Test Results

Null Hypothesis / Direction	Test Statistic	<i>p</i> -Value	Reject H_0 at 5%
Bitcoin $\rightarrow$ HOSE	27.2057	0.0001	Yes
Gold $\rightarrow$ HOSE	5.8952	0.3165	No
Gold and Bitcoin $\rightarrow$ HOSE	32.6265	0.0003	Yes
HOSE $\rightarrow$ Bitcoin	3.9049	0.5632	No
HOSE $\rightarrow$ Gold	2.9509	0.7076	No

Notes: The table reports the Granger causality test results based on the VAR specification. The null hypothesis is H_0 : the excluded variable or block does not Granger-cause the dependent variable. The arrow indicates the direction of predictive causality. Rejection of H_0 at the 5% level suggests statistically significant directional predictability.

Table a7: Weekly VAR Lag Order Selection

Lag Order	AIC	BIC	FPE	HQIC
0	-17.90	-17.88	1.678×10^{-8}	-17.89
1	-18.12	-18.02	1.353×10^{-8}	-18.08
2	-18.18	-18.00	1.271×10^{-8}	-18.11
3	-18.39	-18.14*	1.031×10^{-8}	-18.29*
4	-18.40	-18.07	1.019×10^{-8}	-18.27
5	-18.38	-17.98	1.042×10^{-8}	-18.22
6	-18.38	-17.90	1.039×10^{-8}	-18.19
7	-18.37	-17.82	1.049×10^{-8}	-18.15
8	-18.42*	-17.79	$9.991^* \times 10^{-9}$	-18.17

Notes: The table reports the lag order selection results for the weekly VAR specification. AIC, BIC, FPE, and HQIC denote the Akaike Information Criterion, Bayesian Information Criterion, Final Prediction Error, and Hannan–Quinn Information Criterion, respectively. The asterisk indicates the minimum value for each criterion.

Table a8: Weekly VAR-Based Granger Causality Test Results

Null Hypothesis / Direction	p -Value	Reject H_0 at 5%
Bitcoin $\rightarrow$ HOSE	0.2363	No
Gold $\rightarrow$ HOSE	0.0925	No
Gold and Bitcoin $\rightarrow$ HOSE	0.0927	No
HOSE $\rightarrow$ Bitcoin	0.8444	No
HOSE $\rightarrow$ Gold	0.9421	No

Notes: The table reports the Granger causality test results based on the weekly VAR specification. The null hypothesis is H_0 : the excluded variable or block does not Granger-cause the dependent variable. The arrow indicates the direction of predictive causality. Since none of the null hypotheses is rejected at the 5% significance level, the weekly data provide no strong evidence of directional predictability among the variables.

Table a9: Bitcoin-to-HOSE Granger Causality under Alternative Currency Specifications

Model Specification	Null Hypothesis	Test Statistic	Critical Value	p -Value	df
BTC/USD instead of BTC/VND	Bitcoin $\rightarrow$ HOSE	27.90	11.07	< 0.001	5
BTC/VND with USD/VND control	Bitcoin $\rightarrow$ HOSE	27.66	11.07	< 0.001	5

Notes: The table reports VAR-based Granger causality Wald-test results under alternative currency specifications. The null hypothesis is H_0 : Bitcoin does not Granger-cause the HOSE index. The first specification replaces BTC/VND with the global BTC/USD price, while the second specification includes the USD/VND exchange rate as an exogenous control variable. The reported p -values indicate rejection of the null hypothesis at the 5% significance level in both specifications.

Table a10: Robustness Check of EWMA Correlations under Alternative Decay Factors

λ	Regime	N	HOSE–Gold			HOSE–Bitcoin			Gold–Bitcoin		
			Mean	Median	Neg. Share	Mean	Median	Neg. Share	Mean	Median	Neg. Share
0.90	Normal	1957	-0.0016	0.0092	0.4854	0.0661	0.0774	0.3597	-0.0243	-0.0165	0.5299
0.90	Stress 10%	223	0.0228	0.0341	0.4350	0.0644	0.0596	0.4126	-0.0040	0.0063	0.4888
0.90	Stress 5%	112	0.0240	0.0390	0.4554	0.0775	0.0624	0.4196	0.0130	0.0391	0.4554
0.90	Stress 1%	23	-0.0513	-0.0777	0.5652	0.1153	0.0450	0.4348	-0.0100	0.0693	0.4348
0.94	Normal	1957	-0.0067	-0.0012	0.5023	0.0648	0.0753	0.3316	-0.0264	-0.0160	0.5263
0.94	Stress 10%	223	0.0186	0.0380	0.4439	0.0725	0.0498	0.3453	-0.0037	-0.0040	0.5067
0.94	Stress 5%	112	0.0195	0.0435	0.4464	0.0798	0.0483	0.3661	0.0086	0.0237	0.4286
0.94	Stress 1%	23	-0.0516	-0.0558	0.5652	0.1257	0.0273	0.4348	-0.0091	0.0003	0.4783
0.97	Normal	1957	-0.0083	-0.0072	0.5151	0.0621	0.0573	0.3051	-0.0266	-0.0139	0.5457
0.97	Stress 10%	223	0.0181	0.0366	0.3946	0.0792	0.0641	0.2780	-0.0026	0.0002	0.4978
0.97	Stress 5%	112	0.0169	0.0359	0.3929	0.0901	0.0572	0.2946	0.0058	0.0263	0.4464
0.97	Stress 1%	23	-0.0309	0.0088	0.4348	0.1267	0.0523	0.3478	-0.0063	-0.0013	0.5217

Notes: The table reports EWMA correlation statistics under alternative decay factors, $\lambda = 0.90, 0.94$, and 0.97 . The baseline RiskMetrics specification uses $\lambda = 0.94$. Mean and Median denote the average and median EWMA correlations within each market regime. Neg. Share denotes the proportion of observations in which the corresponding EWMA correlation is negative.

Table a11: Robustness Check of Safe-Haven Total Slopes under Alternative HAC Lag Lengths

HAC Lag	Asset	Stress Regime	Total Slope	Estimate	HAC Std. Error	<i>t</i> -Statistic	<i>p</i> -Value	Significant at 5%
5	Gold	Normal	β_0	0.0440	0.0890	0.4946	0.6209	No
	Gold	Stress 10%	$\beta_0 + \beta_1$	-0.2569	0.1183	-2.1721	0.0298	Yes
	Gold	Stress 5%	$\beta_0 + \beta_1 + \beta_2$	-0.1213	0.1020	-1.1896	0.2342	No
	Gold	Stress 1%	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	-0.1811	0.1909	-0.9487	0.3428	No
	Bitcoin	Normal	β_0	0.1655	0.1331	1.2434	0.2137	No
	Bitcoin	Stress 10%	$\beta_0 + \beta_1$	0.2842	0.2641	1.0759	0.2820	No
	Bitcoin	Stress 5%	$\beta_0 + \beta_1 + \beta_2$	0.0145	0.2079	0.0699	0.9443	No
	Bitcoin	Stress 1%	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	0.9096	0.5056	1.7990	0.0720	No
10	Gold	Normal	β_0	0.0440	0.0873	0.5042	0.6141	No
	Gold	Stress 10%	$\beta_0 + \beta_1$	-0.2569	0.1141	-2.2504	0.0244	Yes
	Gold	Stress 5%	$\beta_0 + \beta_1 + \beta_2$	-0.1213	0.0997	-1.2175	0.2234	No
	Gold	Stress 1%	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	-0.1811	0.1685	-1.0746	0.2825	No
	Bitcoin	Normal	β_0	0.1655	0.1333	1.2421	0.2142	No
	Bitcoin	Stress 10%	$\beta_0 + \beta_1$	0.2842	0.2535	1.1211	0.2622	No
	Bitcoin	Stress 5%	$\beta_0 + \beta_1 + \beta_2$	0.0145	0.2030	0.0716	0.9429	No
	Bitcoin	Stress 1%	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	0.9096	0.4740	1.9189	0.0550	No
20	Gold	Normal	β_0	0.0440	0.0859	0.5121	0.6086	No
	Gold	Stress 10%	$\beta_0 + \beta_1$	-0.2569	0.1091	-2.3548	0.0185	Yes
	Gold	Stress 5%	$\beta_0 + \beta_1 + \beta_2$	-0.1213	0.0980	-1.2383	0.2156	No
	Gold	Stress 1%	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	-0.1811	0.1499	-1.2085	0.2269	No
	Bitcoin	Normal	β_0	0.1655	0.1221	1.3553	0.1753	No
	Bitcoin	Stress 10%	$\beta_0 + \beta_1$	0.2842	0.2517	1.1293	0.2588	No
	Bitcoin	Stress 5%	$\beta_0 + \beta_1 + \beta_2$	0.0145	0.1982	0.0733	0.9416	No
	Bitcoin	Stress 1%	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	0.9096	0.4451	2.0437	0.0410	Yes

Notes: The table reports robustness checks for the safe-haven total slope estimates using Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors with alternative lag lengths of 5, 10, and 20 days. The total slope measures the conditional relationship between each candidate safe-haven asset and the HOSE index under different market stress regimes. A negative and statistically significant total slope is consistent with safe-haven behavior. Significance is evaluated at the 5% level.